# 2026 Level 2 - Fixed Income



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# The Term Structure and Interest Rate Dynamics

a. describe relationships among spot rates, forward rates, yield to maturity, expected and realized returns on bonds, and the shape of the yield curve

b. describe how zero-coupon rates (spot rates) may be obtained from the par curve by bootstrapping

c. describe the assumptions concerning the evolution of spot rates in relation to forward rates implicit in active bond portfolio management

d. describe the strategy of rolling down the yield curve

e. explain the swap rate curve and why and how market participants use it in valuation

f. calculate and interpret the swap spread for a given maturity

g. describe short-term interest rate spreads used to gauge economy-wide credit risk and liquidity risk

h. explain traditional theories of the term structure of interest rates and describe the implications of each theory for forward rates and the shape of the yield curve

i. explain how a bond's exposure to each of the factors driving the yield curve can be measured and how these exposures can be used to manage yield curve risks

j. explain the maturity structure of yield volatilities and their effect on price volatility

k. explain how key economic factors are used to establish a view on benchmark rates, spreads, and yield curve changes

# Term Structure/Interest Rate Dynamics

LOS a, b (10p) Spot Rates, Forward Rates, & the Forward Rate Model – describe

LOS c (4.5p) YTM vs. Spot & Forward Rates – describe

LOS d (5p) Rolling down the YC – describe

LOS e (3.5p) The Swap Rate Curve – explain

LOS f, g (4.5p) Spreads – calculate, interpret, describe

LOS h (5p) Theories of the Term Structure of Interest Rates – explain, describe

LOS i (2.5p) Yield Curve Factor Models – explain

LOS j (4p) Maturity Structure of Yield Volatilities – explain

LOS k (3p) Interest Rate views using Macro Variables – explain

Page 1
LOS a, b
– describe

→ spot rate – rate of interest on a security that makes a single payment at a future date

→ forward rate – interest rate, set today, for a single-payment security to be issued at some future date

$$DF_N = PV = \frac{1}{(1+Z_N)^N}$$

(no coupons) $1 → single unit payment – matures at par

0 ———— N

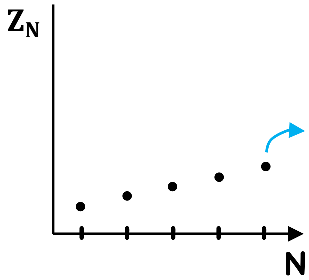
$Z_N$ → spot rate, zero rate, zero-coupon yield

$DF_N$ → discount factor – the price of a default-risk-free, single unit payment in N periods


spot yield curve – the term structure of interest rates

Page 2
LOS a, b
- describe

→ spot curve - the benchmark for the time value of money




annualized return on:
- option-free
- default-risk free
- zero coupon bond

- no reinvestment risk
  ∴ the stated yield ($Z_N$) will be the realized yield if held to maturity

market forces → market price of zeros → $Z_N$ → spot curve
(supply/demand)

∴ shape and level of curve are dynamic

→ forward rate - an interest rate determined today for a loan that will be initiated in a future period
- forward rates are mathematically derived from the spot curve

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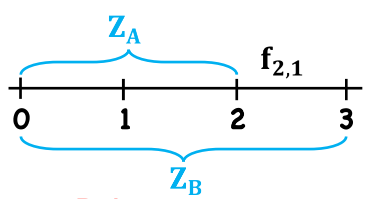
Page 3
LOS a, b
- describe

→ forward pricing model

$$DF_B = DF_A \times F_{A,B-A}$$



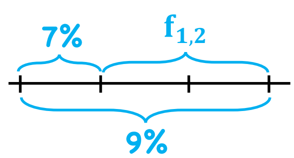

f(when,what)

→ forward rate model

$$(1 + Z_B)^B = (1 + Z_A)^A (1 + f_{A,B-A})^{B-A}$$

- **no arbitrage principle** - securities with identical cash flow payments must have the same price

e.g./ $Z_1$ = 7% , $Z_3$ = 9%, $f_{1,2}$ = ?




pricing: $DF_1 = {}^1/_{1.07} = .9346$

$DF_3 = {}^1/_{(1.09)^3} = .7722$

$DF_3 = DF_1 \bullet F_{1,2}$

$F_{1,2} = {}^{DF_3}/_{DF_1} = {}^{.7722}/_{.9346} = .8262$

longer　　shorter

rate: $(1.09)^3 = (1.07)(1 + f_{1,2})^2$

$1.295029 = 1.07 \bullet (1 + f_{1,2})^2$

$(1 + f_{1,2})^2 = 1.29509/1.07$

$f_{1,2} = (1.29509/1.07)^{1/2} - 1$

$f_{1,2}$ = 10.016%

Page 4
LOS a, b
- describe

**Ex. #2/** $Z_1$ = 9% $Z_2$ = 10% $Z_3$ = 11%

1/ $f_{1,1}$

$$1.1^2 = 1.09(1+f_{1,1})$$

$$f_{1,1} = \left[{(1.1)^2}/{1.09}\right] - 1$$

= 11.01%

2/ $f_{2,1}$

$$(1.11)^3 = (1.1)^2(1+f_{2,1})$$

$$f_{2,1} = \left[{(1.11)^3}/{(1.1)^2}\right] - 1$$

= 13.03%

3/ $f_{1,2}$

$$(1.11)^3 = (1.09)(1+f_{1,2})^2$$

$$f_{1,2} = \left[{(1.11)^3}/{(1.09)}\right]^{1/2} - 1$$

= 12.01%

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$$(1+Z_B)^B = (1+Z_A)^A(1+F_{A,B-A})^{B-A}$$

$$\frac{(1+Z_B)^B}{(1+Z_A)^A} = (1+F_{A,B-A})^{B-A}$$

$$\frac{(1+Z_B)^{B/B-A}}{(1+Z_A)^{A/B-A}} = (1+F_{A,B-A})$$

$$\frac{(1+Z_B)^{\frac{A+B-A}{A-B}}}{(1+Z_A)^{A/B-A}} = (1+F_{A,B-A})$$

$$\frac{(1+Z_B)^{\frac{A}{B-A}+\frac{B-A}{B-A}}}{(1+Z_A)^{A/B-A}} = (1+F_{A,B-A})$$

$$\frac{(1+Z_B)^{A/B-A}}{(1+Z_A)^{A/B-A}}(1+Z_B) = (1+F_{A,B-A})$$

$$\left(\frac{1+Z_B}{1+Z_A}\right)^{A/B-A}(1+Z_B) = (1+F_{A,B-A})$$

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LOS a, b
- describe

$$\left(\frac{1+Z_B}{1+Z_A}\right)^{A/B-A}\left[(1+Z_B) = (1+F_{A,B-A})\right]$$

- if $Z_B > Z_A$, first term > 1, then $F_{A,B-A} > Z_B$

- when the spot curve is upward sloping, the forward curve will lie above the spot curve

- if $Z_B < Z_A$, first term < 1, then $F_{A,B-A} < Z_B$

- when the spot curve is downward sloping, the forward curve will lie below the spot curve

e.g./ $Z_1$ = 9%

$Z_2$ = 10%

$Z_3$ = 11%

is $f_{1,2} > Z_3$?

$Z_1$ $Z_3$

since $Z_3 > Z_1$ → $f_{1,2} > Z_3$

Exh. #2/3

Page 6
LOS a, b
– describe

→ Par curve – YTMs on coupon paying bonds priced at par ($\therefore$ coupon = YTM)

- recently issued on-the-run bonds typically used to construct the par curve (most liquid, typically priced at or near par)

- spot rates can be derived from par rates using a method called bootstrapping

e.g./ 1 yr. = 5% → = 1 yr. spot

2 yr. = 5.97% → $1 = {.0597}/{(1.05)} + {1.0597}/{(1+Z_2)^2}$

3 yr. = 6.91%

4 yr. = 7.81%

$$1 = .05685 + {1.0597}/{(1+Z_2)^2}$$

$$.94314 = {1.0597}/{(1+Z_2)^2}$$

$$(1+Z_2)^2 = {1.0597}/{.94314}$$

$$Z_2 = \left({1.0597}/{.94314}\right)^{1/2} - 1 = 6\%$$

3 yr./ (7%)

$$1 = \frac{.0691}{1.05} + \frac{.0691}{(1.06)^2} + \frac{1.0691}{(1+Z_3)^3}$$

4 yr./ (8%)

$$1 = \frac{.0781}{1.05} + \frac{.0781}{(1.06)^2} + \frac{.0781}{(1.07)^3} + \frac{1.0781}{(1+Z_4)^4}$$

Page 7
LOS a, b
– describe

- relationship between spot rates and one-period forward rates:

$$(1+Z_T)^T = (1+Z_1)(1+f_{1,1})(1+f_{2,1})(1+f_{3,1})\ldots(1+f_{T-1,1})$$

$$Z_T = \underbrace{\{(1+Z_1)(1+f_{1,1})(1+f_{2,1})(1+f_{3,1})\ldots(1+f_{T-1,1})\}^{1/T} - 1}_{\text{geometric mean of } Z_1 \text{ and a series of one period forward rates}}$$

$1

0 1 2 3 4

$$PV_0 = {PV_1}/{(1+Z_1)}$$

$$PV_3 = {1}/{(1+f_{3,1})}$$

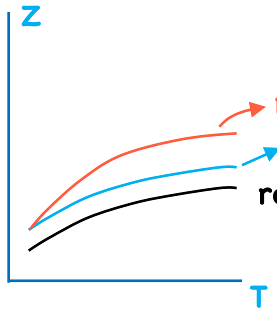
$$PV_2 = {PV_3}/{(1+f_{2,1})}$$

$$PV_1 = {PV_2}/{(1+f_{1,1})}$$

- discounting backwards one period at a time using one-period forward rates

Page 8
LOS a, b
- describe

- yield curves are most commonly upward sloping with diminishing marginal increases



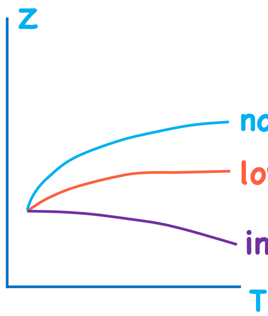

reflects rising long-term inflation expectations

nominal → reflects real rate + expected inflation

real

increases with time

stable




nominal → stable $\pi^e$

lower $\pi^e$

inverted yield curve

expectation of a recession with lower real rates plus expectation of lower inflation

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Page 9
LOS c
- describe

→ Yield-to-Maturity - weighted average of the spot rates used in the valuation of the bond

Ex. #5/ spot rates

$Z_1 = 9\%$

$Z_2 = 10\%$

$Z_3 = 11\%$

YTM = $y_T$

1/ 6%, 2 yr. annual pay, FV = 1000

$$PV = \frac{60}{1.09} + \frac{1060}{(1.1)^2} = 931.08$$

$$931.08 = \frac{60}{(1+y_T)} + \frac{1060}{(1+y_T)^2}$$

→ solve for $Y_T$

PV = -931.08
N = 2
FV = 1000
PMT = 60
$y_T$ = 9.97%

$$Z_1 < y_T < Z_2$$

2/ 5%, 3 y. annual pay, FV = 100

$$PV = \frac{5}{1.09} + \frac{5}{(1.1)^2} + \frac{105}{(1.11)^3} = 85.49$$

$$85.49 = \frac{5}{(1+y_T)} + \frac{5}{(1+y_T)^2} + \frac{105}{(1+y_T)^3}$$

PV = -85.49
N = 3
PMT = 5
FV = 100

$y_T$ = 10.93%

$$Z_1 < Z_2 < y_T < Z_3$$

YTM = mwrr = bond's IRR

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LOS c
- describe

→ YTM is the expected rate of return on a bond if:

1) held to maturity
2) all coupon/principal payments made in full
3) coupons are reinvested at the YTM rate (if they are reinvested)

typically does not hold

e.g./ N = 3, PMT = 5, FV = 100, PV = 85.49

$y_T$ = 10.93% → $85.49(1.1093)^3 = 116.697$

vs. $\underbrace{5(1.1093)^2 + 5(1.1093)}_{\text{reinvestment of coupon}} + 105 = 116.697$

YTM is a poor proxy for E(R) if:

1) interest rates are volatile
2) yield curve is not flat } reinvestment rate ≠ YTM
3) significant risk of default → cash flows differ
4) embedded options → holding period may be different

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LOS c
- describe

e.g./ 5 yr., 10% annual pay

$Z_1 = 5\%$
$Z_2 = 6\%$
$Z_3 = 7\%$
$Z_4 = 8\%$
$Z_5 = 9\%$

$$PV = \frac{10}{1.05} + \frac{10}{(1.06)^2} + \frac{10}{(1.07)^3} + \frac{10}{(1.08)^4} + \frac{110}{(1.09)^5} = 105.43$$

YTM/ PV = -105.43 N = 5 PMT = 10 FV = 100

CPT $^I/_Y$ = 8.6178%

$105.43(1.086178)^5 = 159.393$

$\left.\begin{array}{l} f_{1,1} = 7\% \\ f_{2,1} = 9\% \\ f_{3,1} = 11.1\% \\ f_{4,1} = 13.1\% \end{array}\right\}$ expected future spot rates →

| | |
|---|---|
| 10(1.07)(1.09)(1.111)(1.131) | = 14.655 |
| 10(1.09)(1.111)(1.131) | = 13.696 |
| 10(1.111)(1.131) | = 12.565 |
| 10(1.131) | = 11.310 |
| 110 | = 110 |
| | 162.2264 |

$$E(R) = \left(^{162.2264}/_{105.43}\right)^{1/5} - 1 = \underline{9\%}$$

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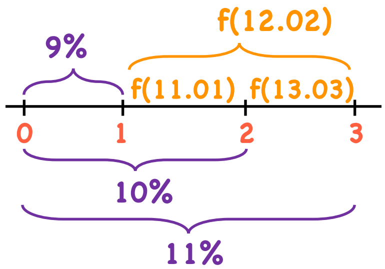
LOS d
- describe

$DF_1 = .9174$

$DF_2 = .82644$

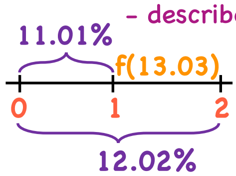
$DF_3 = .73119$




after → one → year

$DF_1 = .900819$

$DF_2 = .79699$




1 yr. holding period → all bonds return $Z_1$ if spot curve evolves to forward curve

**After 1 year:**

1 yr. zero: $\left({}^{1}\!/_{.9174}\right) - 1 = 9\%$

2 yr. zero: $\left({}^{1}\!/_{1.1101}\right) = .900819 \longrightarrow \left({}^{.900819}\!/_{.82644}\right) - 1 = 9\%$

or .82644(1.09) = .900819

3 yr. zero: $\dfrac{1}{(1.1303)(1.1101)} = .79699 \longrightarrow \left({}^{.79699}\!/_{.73119}\right) - 1 = 9\%$

or .73119(1.09) = .79699

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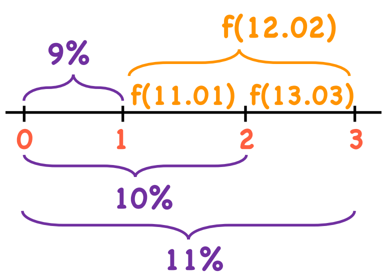
LOS d
- describe

$DF_1 = .9174$

$DF_2 = .82644$

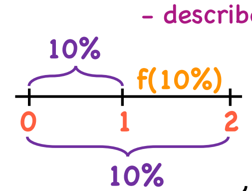
$DF_3 = .73119$




after → one → year

$DF_1 = .90909$

$DF_2 = .82644$




- rates rise, but not to the rates implied by the forward curve

**After 1 year:**

1 yr. zero: $\left({}^{1}\!/_{.9174}\right) - 1 = 9\%$

2 yr. zero: $\left({}^{.90909}\!/_{.82644}\right) - 1 = 10\%$

3 yr. zero: $\left({}^{.82644}\!/_{.73119}\right) - 1 = 13.027$

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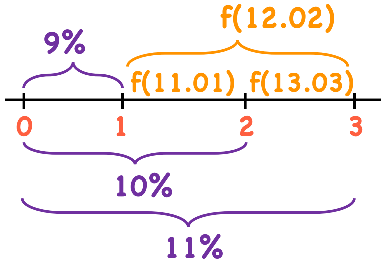
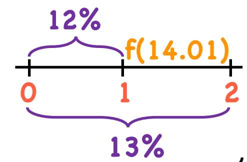
LOS d
- describe

$DF_1$ = .9174
$DF_2$ = .82644
$DF_3$ = .73119




after → one → year

$DF_1$ = .89286
$DF_2$ = .78315




- rates rise above the rates implied by the forward curve

After 1 year:

1 yr. zero: $\left(^{1}/_{.9174}\right) - 1 = 9\%$

2 yr. zero: $\left(^{.89268}/_{.82644}\right) - 1 = 8.037\%$

3 yr. zero: $\left(^{.78315}/_{.73119}\right) - 1 = 7.106\%$

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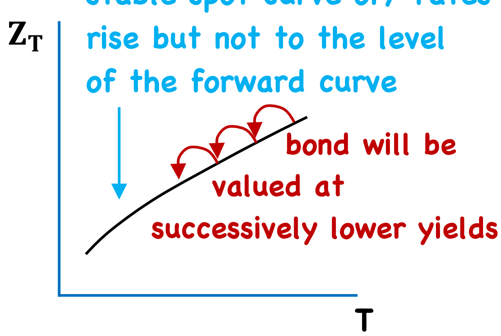
LOS d
- describe

| After 1 year | spot evolves to the forward curve | spot rises but not to the forward curve | spot rises above the forward curve |
|---|---|---|---|
| 1 yr. | 9% matured | 9% matured | 9% matured |
| 2 yr. | 9% | 10% | 8.037% |
| 3 yr. | 9% | 13.027% | 7.106% |
| | | buy longer than IH | IH matching |

stable spot curve or/ rates rise but not to the level of the forward curve




→ Riding the yield curve, also called 'rolling down the yield curve'

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LOS d
- describe

- upward sloping spot curve, forward lies above

- if the level and shape of the spot curve is not expected to change, or not expected to rise to the level of the forward curve, then:

  - buying bonds with a maturity longer than the investment horizon would provide a total return greater than the return on a $\underbrace{\text{maturity-matching}}_{\text{with the IH}}$ strategy

- the $\underbrace{\text{wider the spread}}_{\text{Steeper spot curves = wider spreads}}$, the longer the bond, the greater the total return

E(R) = coupons + reinvestment of coupon $^{+}/_{-}$ capital gain/loss on sale

Page 17
LOS e
- explain

→ Swap Rate Curve/

- swap rate → the rate on the fixed leg of an interest rate swap
  → derived using short-term lending rates rather than default-risk-free rates

- swap rates were based on Libor - a survey-based rate for an unsecured loan
  - Libor to be replaced by a market-referenced rate (MRR) based on secured overnight funding transactions

- the yield curve of swap rates is called the swap rate curve (swap curve)
  - swap rates are a type of par rate

  - swaps are priced such that $PV_{fl} = PV_{fx}$
    - since $PV_{fl} = 1$, $PV_{fx} = 1$

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LOS e
- explain

- swap market is highly liquid
  - for countries lacking a liquid gov't. bond market > 1 yr., swap curve is the benchmark for interest rates
  - for countries where private sector > public sector, swap curve is a more relevant measure of time value
- both spot and swap curves are used in FI valuation
  - depends on the relative liquidity of both and the interest rate exposure profile of the institution using the curve
- Swap rate/ $1 = \frac{PMT}{(1+Z_1)} + \frac{PMT}{(1+Z_2)^2} + \cdots + \frac{PMT}{(1+Z_N)^N} + \frac{FV}{(1+Z_N)^N}$

$$1 = PMT\left(\frac{1}{(1+Z_1)}\right) + PMT\left(\frac{1}{(1+Z_2)^2}\right) + \cdots + PMT\frac{1}{(1+Z_N)^N} + FV\left(\frac{1}{(1+Z_N)^N}\right)$$

$$1 = PMT \bullet DF_1 + PMT \bullet DF_2 + \cdots + PMT \bullet DF_N + FV \bullet DF_N$$

$$1 = PMT(\sum DF_N) + DF_N$$

$$1 - DF_N = PMT(\sum DF_N) \rightarrow \boxed{PMT = \frac{1-DF_N}{\sum DF}}$$

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LOS e
- explain

ex. #8/

$$PMT = \frac{1-DF_N}{\sum DF}$$

$DF_1 = .9524$

$DF_2 = .8900$

$DF_3 = .8163$

$DF_4 = .7350$

$$Z_1 = \frac{1-.9524}{.9524} = 5\% \qquad Z_2 = \frac{1-.8900}{1.8424} = 5.97\%$$

$$Z_3 = \frac{1-.8163}{2.6587} = 6.909\% \qquad Z_4 = \frac{1-.7350}{3.3937} = 7.808\%$$

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LOS f
- calculate
- interpret

- swap spread - the spread paid by the fixed-rate payer of an interest rate swap over the rate of the 'on-the-run' gov't. security with the same maturity as the swap

e.g./ 5-yr. swap rate = 2.00%
5-yr. Treasury = 1.70%
swap spread = 30bps

Swap rate
30bps — swap spread — credit, liquidity
1.70% — TVM component

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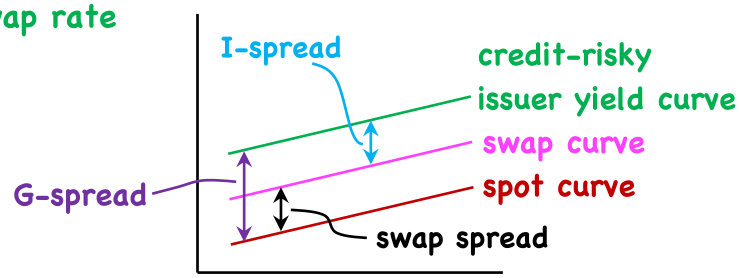
LOS g
- describe

- a measure of a specific bond's credit and liquidity risk is the Z-spread
  - a constant basis point spread added to each spot rate (or swap rate - depending of how a bond is quoted) so that DCF of a bond = current price

$$PV = \frac{PMT}{(1 + r_1 + Z)} + \frac{PMT}{(1 + r_2 + Z)^2} + \cdots + \frac{PMT + FV}{(1 + r_N + Z)^N}$$

→ I-spread → spread between a bond's YTM and an equal maturity swap rate

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LOS g
- describe

→ TED spread → 3-mos. USD Libor - 3-mos. T-Bill rate
- typically ranges 10 - 50 bps
- rising RED spread indicates liquidity is being withdrawn
∴ good barometer of credit and liquidity risk in the general economy

→ Libor-OIS spread (e.g. 3-month Libor - 3-mos. OIS rate) - typically ~10bps
- rec. a fixed OIS term rate, pay a floating rate that is:

overnight indexed swap rate

FFR

$$[(1 + day_1)(1 + day_2) \ldots (1 + day_t)]^{1/t}$$ - geometric average

daily unsecured overnight rate (between banks)

days in the payment period

→ SOFR - secured overnight financing rate → market determined, collateralized by UST

daily, volume-weighted index of repo. transactions, influenced by supply/demand for secured overnight funding

Page 22
LOS h
- explain
- describe

→ Expectations Theory/

• Pure expectations theory (unbiased expectations theory)

- forward rates are unbiased predictors of future spot rates

∴ bonds of any maturity are perfect substitutes

e.g./ E(R): 3 yr. bond = 3 x 1 yr. bonds = buy 5-yr., sell in 3 yrs.

- assumes investor risk neutrality (uncertainty = risk)
- contradicts evidence of investor risk aversion

• Local expectations theory

E(R) ∀ bonds = $r_f$ over short time periods but not over longer periods

- however, 90-day Treasury demand > 30-year Tr. demand

∴ 90-day returns < 90-day returns on 30 yr. bonds

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LOS h
- explain
- describe

→ Liquidity Preference Theory/ • LPs exist → compensate for interest rate risk when lending long term → increase with maturity

- liquidity in this sense means having to sell the bond at some uncertain price (not lack of a market)

∴ predicts an upward sloping YC → forward rates, as an expectation of future spot rates, are biased upward by the LP

→ Segmented Markets Theory/ each maturity segment can be thought of as a segmented market in which yields are determined independently from the yields that prevail in other maturity segments

∴ forward rates are not an expectation of future spot rates or liquidity premiums

• market participants are unwilling/unable to invest in anything other than securities of their preferred maturity

Page 24
LOS h
- explain
- describe

→ Preferred Habitat/ • borrowers and lenders have a preference for particular maturities, but yields at different maturities are not determined independently

- if returns/costs are compelling in other maturities, preferences will follow

- retains the moderate parts of expectations theory and segmented markets theory

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LOS i
- explain

→ Yield curve factor models/

• Shaping risk - the sensitivity of a bond's price to the changing shape of the YC

- for active bond mgmt. → base trades on a forecasted YC shape
→ hedge YC risk on a bond portfolio

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Page 25
LOS i
- explain

→ Yield curve factor models/

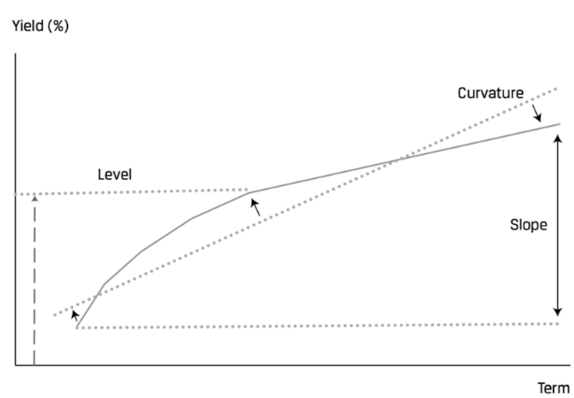
- factors affecting the shape of the YC

Litterman & Scheinkman (1991) → 3-factor model
- level (up, down)
- steepness (slope)
- curvature

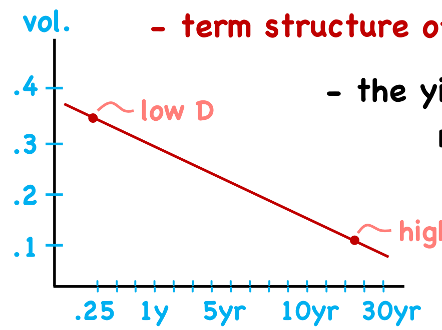
Level - dominant effect - explains most of the total changes in yields

Steepness - S.t. rates moving more than L.T. rates
- explain less than 'Level'

Curvature → short/long falling, intermediate unch./rising
- neg. for mid-curve bonds
- smallest impact of all three factors



Page 26
LOS j
- explain




- term structure of interest rate volatility

- the yield volatility for a zero at each maturity

- price volatility will depend on duration

100 bps x 2 < 20 bps x 18
(D) (D)

| Maturity (years) | 0.25 | 0.50 | 1 | 2 | 3 | 5 | 7 | 10 | 20 | 30 |
|---|---|---|---|---|---|---|---|---|---|---|
| $\sigma(t,T)$ | 0.3515 | 0.3173 | 0.2964 | 0.2713 | 0.2577 | 0.2154 | 0.1885 | 0.1621 | 0.1332 | 0.1169 |

annualized

$\underbrace{0.25 \quad 0.50 \quad 1}_{\text{more volatile}}$

more volatile
- more strongly linked to uncertainty regarding monetary policy

$\underbrace{5 \quad 7 \quad 10 \quad 20 \quad 30}_{\text{more strongly linked to uncertainty regarding the real economy and inflation}}$

more strongly linked to uncertainty regarding the real economy and inflation

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LOS j
- explain

→ Managing $\underbrace{\text{YC Risk}}$

risk to a portfolio from unanticipated changes in YC

a) EffDur → parallel changes only

allow for mgmt. of shaping risk

b) key rate duration - sensitivity to a rate change at a specific maturity segment

c) factor sensitivities

e.g./

| | $100 | $100 | $100 |
|---|---|---|---|
| **Year** zeros → | **1** | **5** | **10** |
| Parallel | 1 | 1 | 1 |
| Steepness | −1 | 0 | 1 |
| Curvature | 1 | 0 | 1 |
| Duration = | 1 | 5 | 10 |

Portfolio value = $300

Portfolio duration

$= {}^1/_3\,(1) + {}^1/_3\,(5) + {}^1/_3\,(10)$

$= 5.333$

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LOS j
- explain

| Year | Zeros → \$100 1 | \$100 5 | \$100 10 |
|---|---|---|---|
| Parallel | 1 | 1 | 1 |
| Steepness | –1 | 0 | 1 |
| Curvature | 1 | 0 | 1 |
| Duration = | 1 | 5 | 10 |

$$KRD = \frac{D}{P_0 \bullet \Delta_y} \longrightarrow \frac{1}{300(.01)} + \frac{5}{300(.01)} + \frac{10}{300(.01)}$$

.3333 + 1.6667 + 3.3333 = 5.333

$$\therefore KRD_{full} = -.3333 \bullet \Delta Z_1 - 1.6667\Delta Z_5 - 3.3333\Delta Z_{10}$$

$$KRD_L = KRD_{full} = 5.3333 \qquad KRD_S = \frac{(D_1 + D_5 + D_{10})}{300(.01)} = \frac{-1+10}{3} = {}^9/_3 = 3$$

$$KRD_C = \frac{(D_1 + D_5 + D_{10})}{300(.01)} = \frac{1+10}{3} = {}^{11}/_3 = 3.6667$$

$$KRD_{full} = -5.333\Delta X_L - 3\Delta X_S - 3.6667\Delta X_C$$

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LOS j
- explain

$$KRD_{full} = -5.333\Delta X_L - 3\Delta X_S - 3.6667\Delta X_C$$

e.g./ $\Delta X_L$ = -.0050 $\Delta X_S$ = 0.002 $\Delta X_C$ = 0.001

$$\%\Delta PV_{full} = \underbrace{-5.333(-.0050)}_{\text{Level}} - \underbrace{3(.002)}_{\text{Slope}} - \underbrace{3.6667(.001)}_{\text{curvature}} = 1.7\%$$

---

LOS k
- explain

$\underbrace{\text{short} \qquad \text{intermediate}}$ long

${}^1/_3$ → inflation, GDP growth

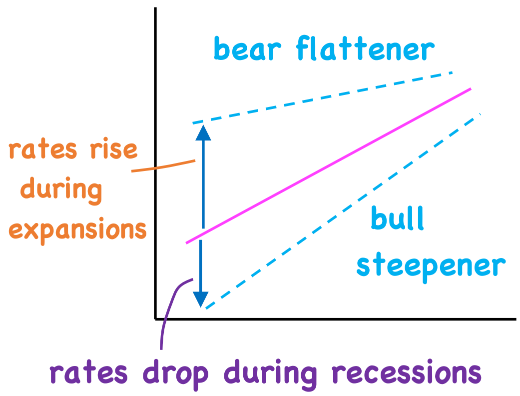
${}^2/_3$ → monetary policy

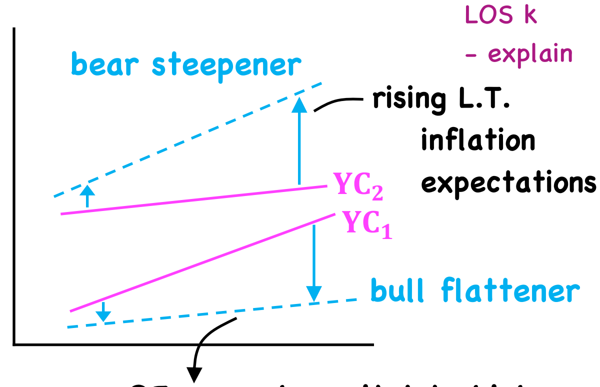
long ↓ monetary policy ← ${}^1/_3$

inflation ← ${}^2/_3$

factors explaining variance of bond yields

Page 30
LOS k
- explain




QE → push capital to higher risk assets

or/ flight to quality → capital looks for lower risk assets

→ Fiscal policy/ affects bond supply and yields

- larger/longer deficits = more supply = higher yields

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LOS k
- explain

→ Maturity Structure - more/less supply of any maturity segment (e.g. 30 yr.) will lower/raise yields at that segment

→ Investor demand (domestic/non-domestic) - raises prices, lowers yield

• Active mgmt./

rates expected to:

↓ - extend D

↑ - shorten D

curve expected to:

flatten → barbell structure
ow s.t./L.T.
uw intermediate

Steepen → bullet structure
ow intermediate
uw s.t./L.T.

# The Arbitrage-Free Valuation Framework

a. explain what is meant by arbitrage-free valuation of a fixed-income instrument

b. calculate the arbitrage-free value of an option-free, fixed-rate coupon bond

c. describe a binomial interest rate tree framework

d. describe the process of calibrating a binomial interest rate tree to match a specific term structure

e. describe the backward induction valuation methodology and calculate the value of a fixed-income instrument given its cash flow at each node

f. compare pricing using the zero-coupon yield curve with pricing using an arbitrage-free binomial lattice

g. describe pathwise valuation in a binomial interest rate framework and calculate the value of a fixed-income instrument given its cash flows along each path

h. describe a Monte Carlo forward-rate simulation and its application

i. describe term structure models and how they are used

# Arbitrage-Free Valuation

Page 1
LOS a
- explain

- arbitrage - riskless profit with zero inv.
- principle of no arbitrage - prices adjust until there are no arbitrage opportunities
- Law of one Price/ - two goods that are perfect substitutes must sell for the same current price (in the absence of transaction costs)
- Arbitrage Opportunity/
  ① Value Additivity - the value of the whole should equal the sum of the value of the parts

  Asset A .952381 ⟶ 1
  B $95 ⟶ 105 (Portfolio of 105 units of A)

  Sell 105 units of A (105 x .952381 = 100), Buy asset B
  - generates $5 today, $0 at maturity

Page 2
LOS a
- explain

- Arbitrage Opportunity/
  ② Dominance

  $\left.\begin{array}{ll} C & \$100 \longrightarrow 105 \\ D & \$200 \longrightarrow 220 \end{array}\right\}$ both risk-free

  - Sell 2 x C = 200 ⟶ -210
  - Buy D = 200 ⟶ +220

  Ø +10

- Arbitrage-Free Valuation/ an approach to security valuation that determines security values that are consistent with the absence of arbitrage opportunities

Page 3
LOS b
- calculate

e.g./ Bond A ⇒ 3% - 10 yr. annual, YTM = 2.5% in New York
- sells for 104.376 in Chicago

N = 10, PMT = 3 , $^{I}/_{Y}$ = 2.5 FV = 100 CPT PV = 104.376

Bond B ⇒ 3% - 10 yr. annual, YTM = 3.2% in Hong Kong
- sells for 97.22 in Shanghai

N = 10 PMT = 3 $^{I}/_{Y}$ = 3.2 FV = 100 PV = 98.311

⇒ Buy in Shanghai
⇒ Sell in Hong Kong

Page 4
LOS b
- calculate

e.g./ Par rates
1 yr. 2%
2 yr. 3%
3 yr. 4%

3-yr., 5% annual @ 102.7751
YTM = 4%
- is this an arbitrage-free value?

- traditional valuation framework
  - estimate cash flows
  - select a discount rate
  - compute PV

Spot rates/

$$100 = \frac{102}{1 + r(1)}, r(1) = 2\%$$

$$100 = \frac{3}{1.02} + \frac{103}{(1 + r(2))^2}; r(2) = 3.015\%$$

$$100 = \frac{4}{1.02} + \frac{4}{1.03015^2} + \frac{104}{(1 + r(3))^3}; r(3) = 4.055\%$$

$$PV = \frac{5}{(1.02)} + \frac{5}{(1.03015)^2} + \frac{105}{(1.04055)^3}$$

$$= 102.8102$$

- will not work with bonds with embedded options/

# Binomial Interest Rate Tree

Page 5
LOS c
- describe

- for option-free bonds, DCF w/ spot rates
= arb-free valuation
but/ not so with embedded options

⇒ bonds with embedded options have cash flows that are interest rate dependent
(e.g. Callable bond, as r ↓, prob. of call ↑)

- Need an approach that derives an arb-free valuation for both option-free and embedded option bonds

∴ valuation framework must allow interest rates to take on different potential values in the future based on some assumed level of volatility

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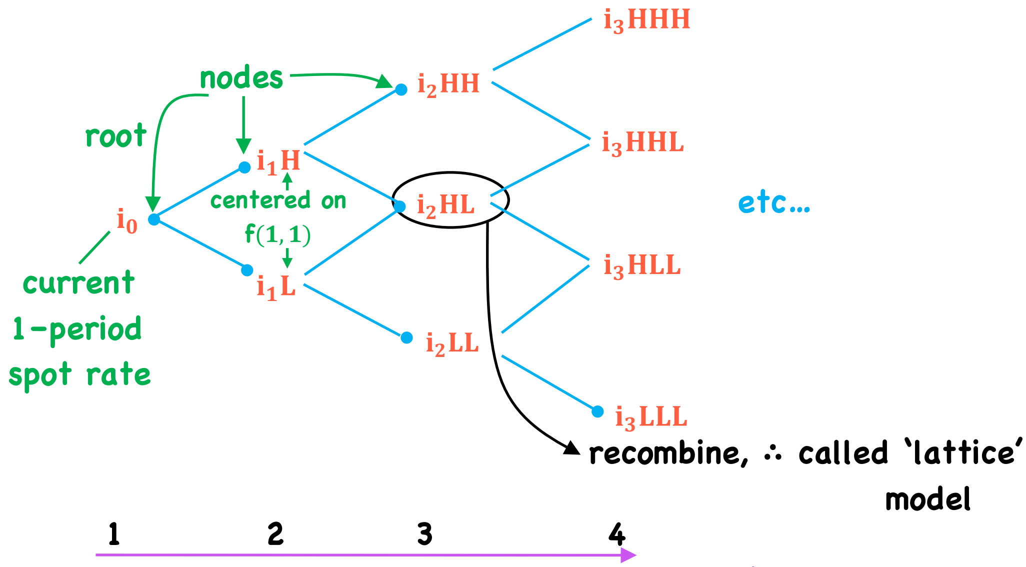
LOS c
- describe




only 1 rate being modelled
- one factor model

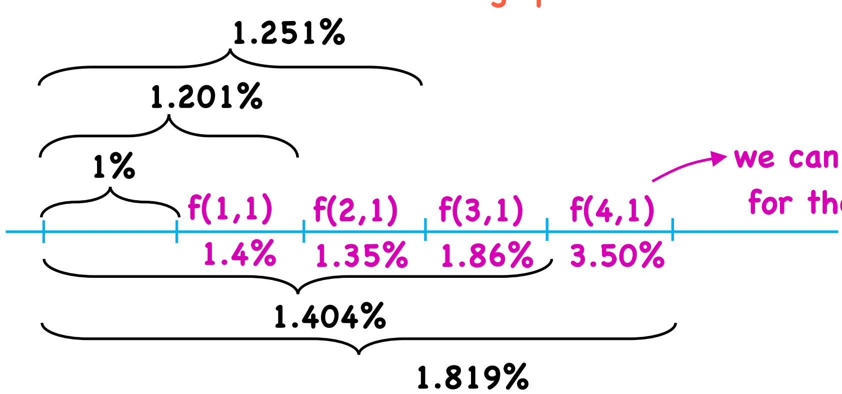
goal is to populate the tree w/ interest rates (under constraints)

Page 7
LOS c
- describe

- we observe the following par rates/

| | | | |
|---|---|---|---|
| 1 yr. | 1% | PV = 100 | 1% |
| 2 yr. | 1.2% | | 1.201% |
| 3 yr. | 1.25% | | 1.251% |
| 4 yr. | 1.4% | | 1.404% |
| 5 yr. | 1.8% | | 1.819% |

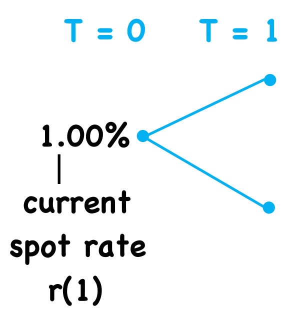
- we can uncover the following spot rates




we can then solve for the forward rates

Page 8
LOS c
- describe

- generate an interest rate tree




- how do we move from T = 0 to T = 1?
- need ① assumption about volatility
  - historical
  - implied - based on observed prices of int. rate derivatives

  ② an interest rate model
  - lognormal random walk

random vars. that only take positive values

∴ no neg. rates
+
higher volatility at higher rates

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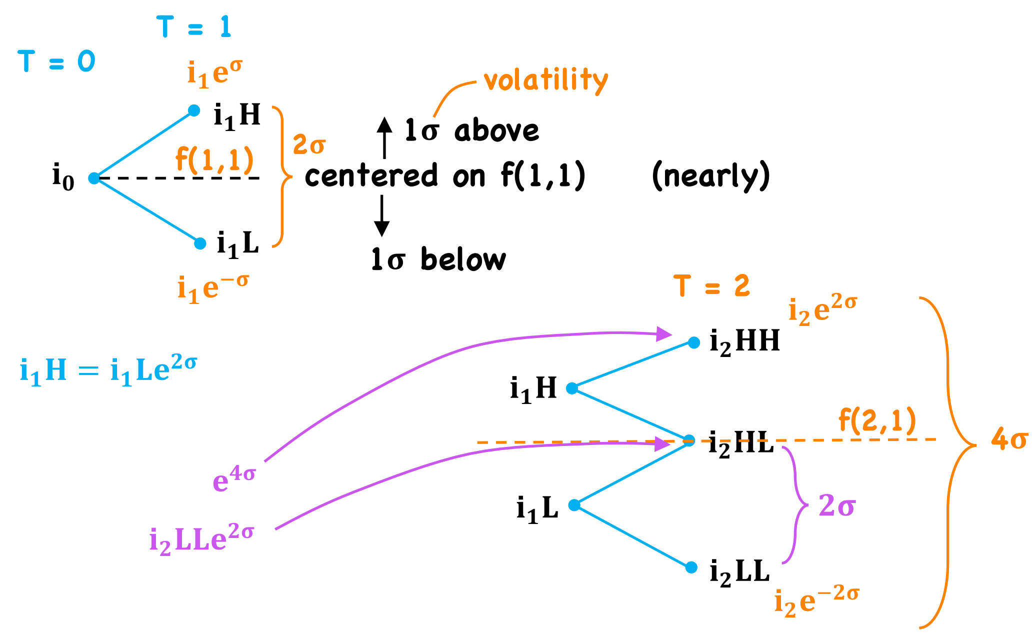
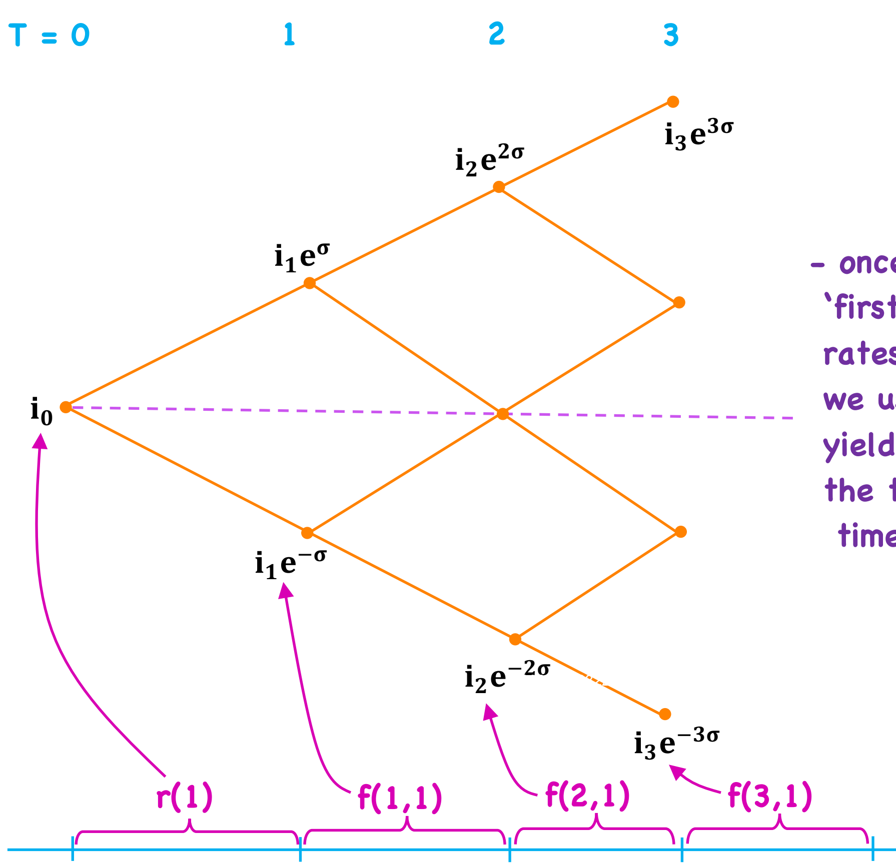
LOS c
- describe




Page 10
LOS c
- describe

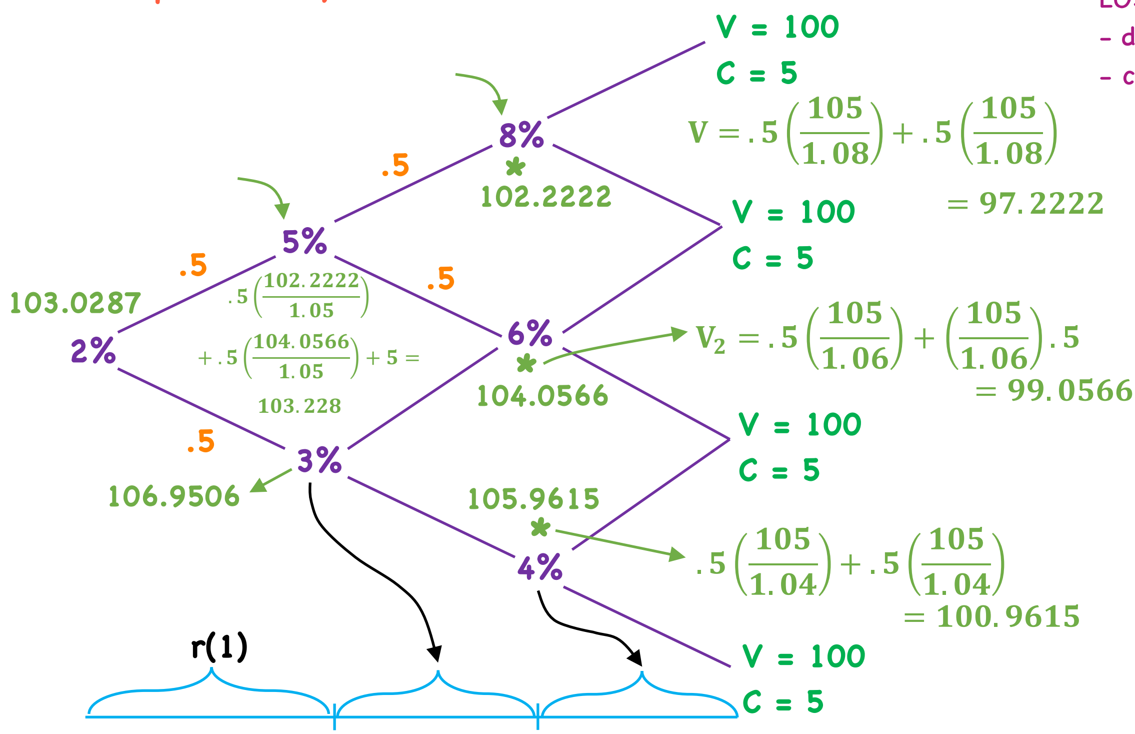



- once all the 'first-pass' interest rates are populated, we use par yields to calibrate the tree, one time-step at a time

## Backward Induction

- find price of 3 yr., annual 5% bond

Page 11
LOS d, e
- describe
- calculate




## Calibration

2-yr. par = 1.2%     f(1,1) = 1.4%     r(1) = 1%
σ = .15

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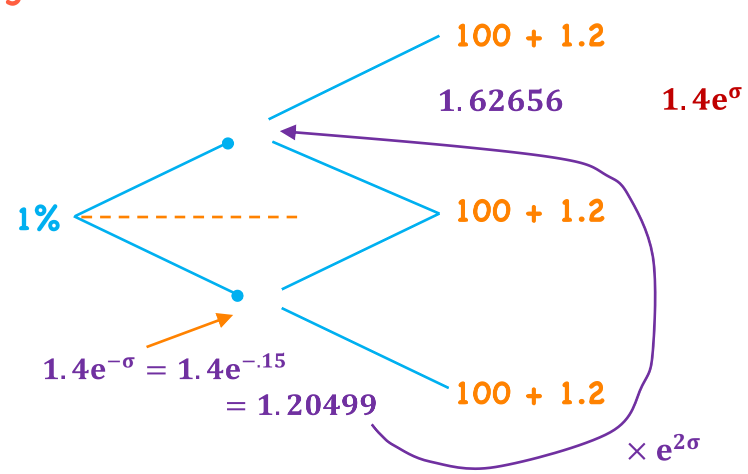
LOS d, e
- describe




1.4σ = 1.4(.15) = 21 bps     linear

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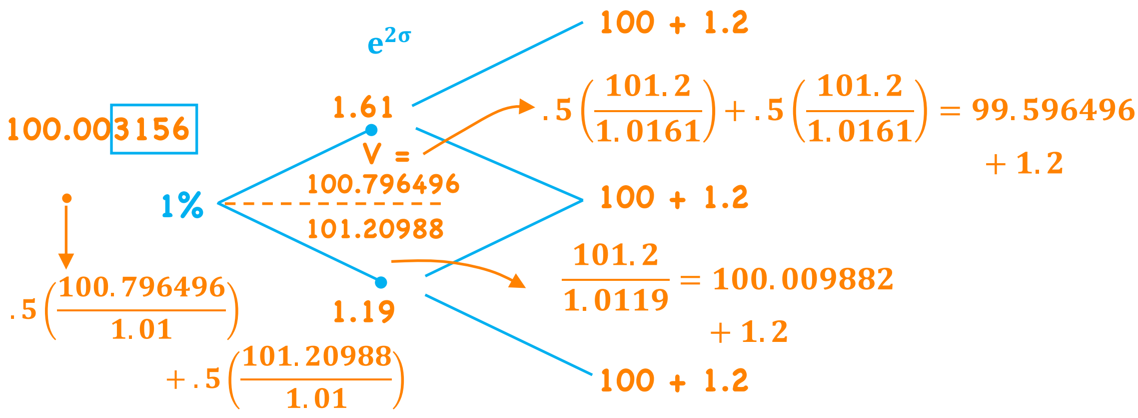
LOS d, e
- describe

2-yr. par = 1.2%  $f(1,1) = 1.4\%$  $r(1) = 1\%$

$\sigma = .15$




$1.4\sigma = 1.4(.15) = 21$ bps

∴ 1.19 too low, next iteration
1.192

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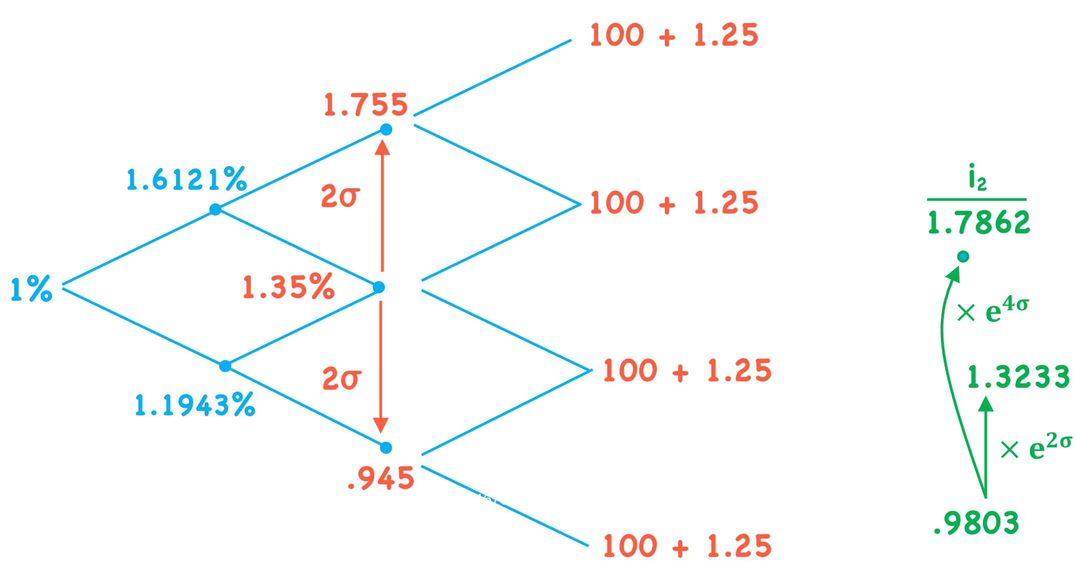
LOS d, e
- describe

3-yr. par = 1.25%  $f(2,1) = 1.35\%$  $r(1) = 1\%$

$\sigma = .15$  $i_1H = 1.6121\%$  $i_1L = 1.1943\%$




$f(2,1)\sigma = 1.35(.15) = 20.25$bps

$i_2LL = f(2,1) - 2\sigma = 1.35 - 2(20.25\text{bps})$

$= .945$

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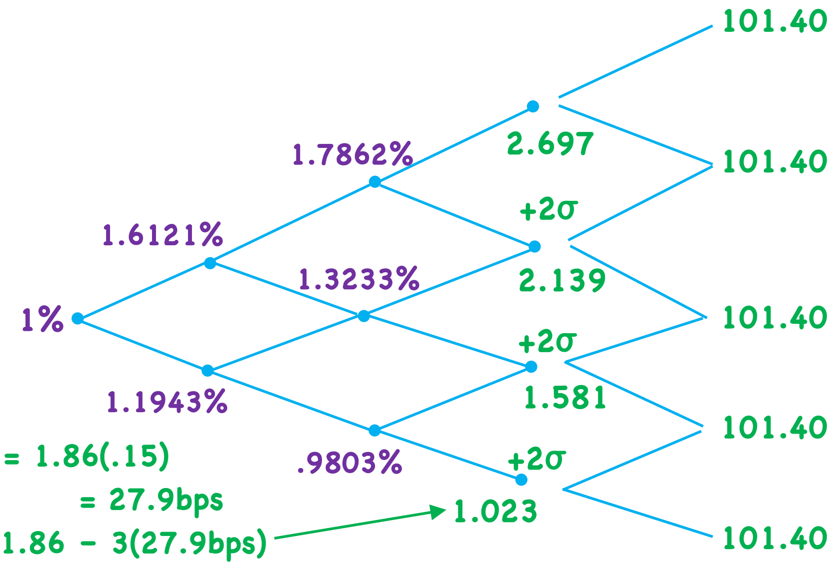
LOS d, e
– describe

4-yr. par = 1.4%  f(3,1) = 1.86%  r(1) = 1%

σ = .15  $i_1H$ = 1.6121%  $i_1L$ = 1.1943%

$i_2HH$ = 1.7862%  $i_2HL$ = 1.3233%  $i_2LL$ = 0.9803




f(3,1)σ = 1.86(.15)
= 27.9bps
$i_3LLL$ = 1.86 – 3(27.9bps) → 1.023

<u>actual</u>
2.8338
× $e^{6\sigma}$
2.0994
× $e^{4\sigma}$
1.5552
× $e^{2\sigma}$
1.1521

## Calibrate & Confirm

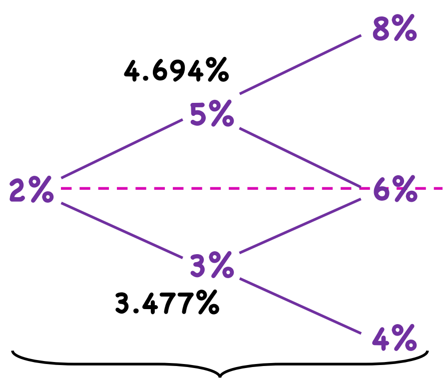
Page 15
LOS f
– compare

| <u>Par rate</u> | | <u>Spot</u> | <u>P(T)</u> |
|---|---|---|---|
| 1 yr. | 2% | 2% | .9804 |
| 2 yr. | 3% | 3.015% | .9423 |
| 3 yr. | 4% | 4.055% | .8876 |

$\sigma = 15\%$

$$\frac{100}{(1.03015)^2}$$

$$\frac{100}{(1+r(1))(1+f(1,1))}$$




2 yr. spot = .9423

using first pass rate → .9419
∴ too high

Root Node/ $Y_0 = S_0 = F_0$ = 2%
par = spot = forward

T = 1 nodes/

$$f(1,1) = \frac{(1.03015)^2}{1.02} - 1 = 4.040\%$$

$$f(1,1d) = 4.040e^{-.15} = 3.477\%$$

$$f(1,1u) = 4.040e^{.15} = 4.694\%$$

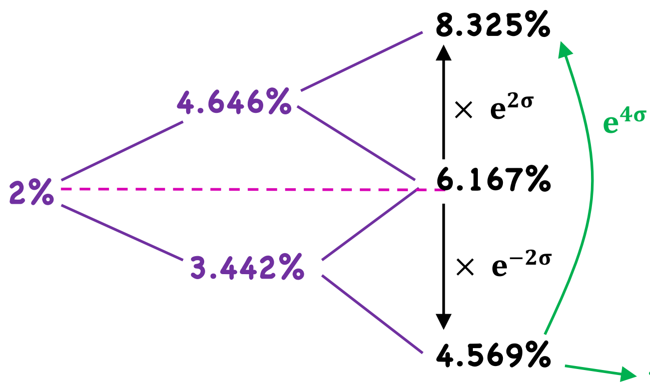
} first best guess

× $e^{2\sigma}$
(3.442%, 4.646%)

Page 16
LOS f
- compare

| | Par rate | Spot | P(T) |
|---|---|---|---|
| 1 yr. | 2% | 2% | .9804 |
| 2 yr. | 3% | 3.015% | .9423 |
| 3 yr. | 4% | 4.055% | .8876 |

$\sigma = 15\%$




too high

T = 2 nodes/

$$f(2,1) = \frac{(1.04055)^3}{(1.03015)^2} - 1 = 6.167\%$$

Using 3 yr.-spot @ .8876

- tree produces .8866

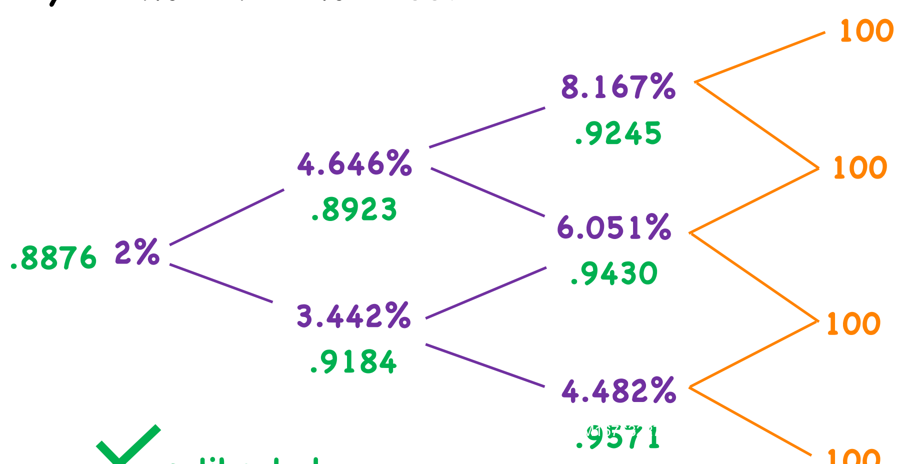
Actual: 4.482%, 6.051%, 8.167%

Page 17
LOS f
- compare

| | Par rate | Spot | P(T) |
|---|---|---|---|
| 1 yr. | 2% | 2% | .9804 |
| 2 yr. | 3% | 3.015% | .9423 |
| 3 yr. | 4% | 4.055% | .8876 |

$\sigma = 15\%$

Use a 3-yr. Zero




calibrated

3-yr. annual, 5% coupon @ 102.8102 YTM = 4

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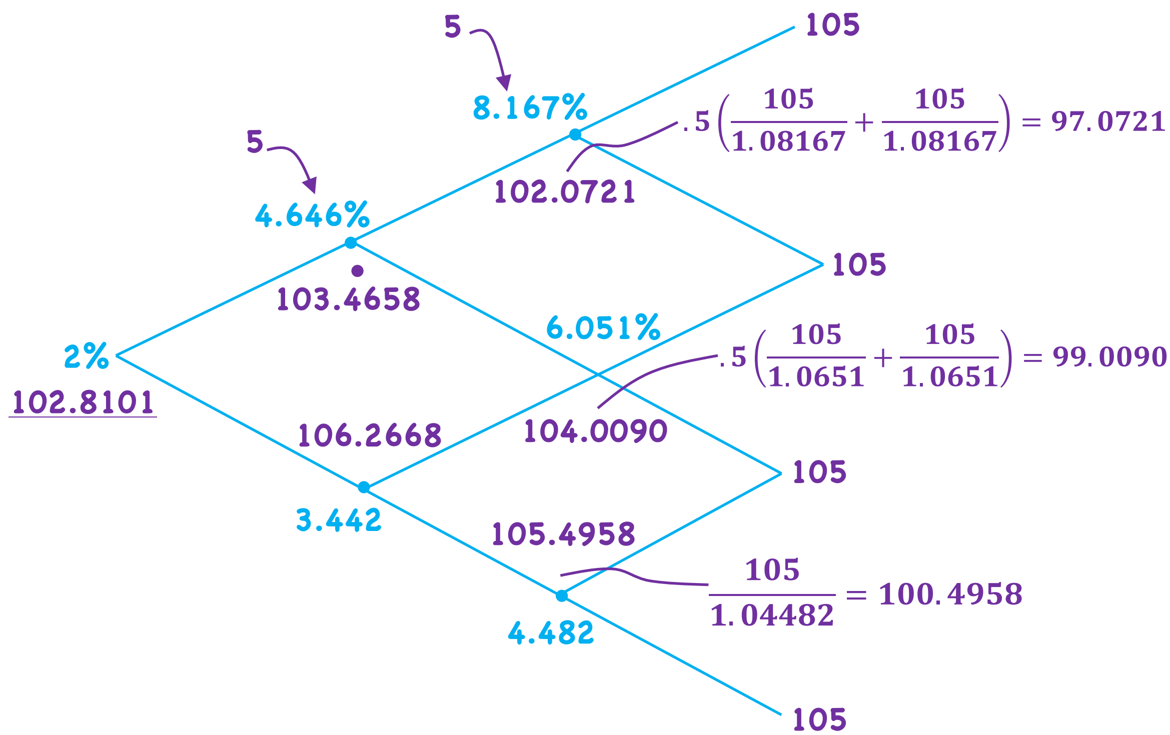
LOS f
- compare




## Pathwise Valuation

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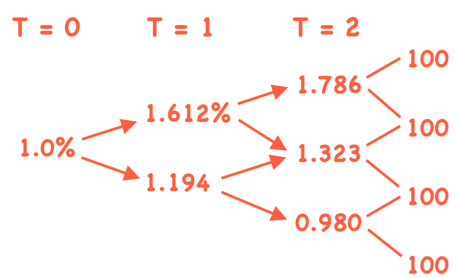
LOS g
- describe
- calculate

- an alternative to backward induction in a binomial tree
- calculate PV of a bond for each possible interest rate path then take the average

Step # 1/ Count the # of paths
- use a triangular array of binomial coefficients

| Discount a/ | | |
|---|---|---|
| 1 yr. | 1 | T = 0 |
| 2 yr. | 1 1 | T = 1 |
| 3 yr. | 1 2 1 | T = 2 |
| 4 yr. | 1 3 3 1 | T = 3 |
| 5 yr. | 1 4 6 4 1 | T = 4 |
| 6 yr. | 1 5 10 10 5 1 | T = 5 |



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LOS g
- describe
- calculate


3 yr. zero

r(3) = 1.251

$P(3) = \frac{100}{(1.01251)^3} = 96.3389$

| Path | | | | PV |
|---|---|---|---|---|
| 1 | 1.0 | 1.612 | 1.786 | 100/(1.01786)(1.01612)(1.01) = 95.7294 |
| 2 | 1.0 | 1.612 | 1.323 | 100/(1.01323)(1.01612)(1.01) = 96.1670 |
| 3 | (1.0) | (1.194) | (1.323) | 100/ = 96.5638 |
| 4 | (1.0) | (1.194) | (0.980) | 100/ = 96.8915 |

$\Sigma/_4$ = 96.3379

## Monte Carlo Simulation

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LOS h
- describe

⇒ Take a 30-yr. semi – how many paths?

- MC is a method for simulating a sufficiently large number of interest rate paths

  - paths are generated based on some probability distribution and some assumption about volatility
  - fit to the current spot curve

    i.e. avg. of all paths for each benchmark bond equals its actual market value

    ∴ renders it arb-free

⇒ most often used when CFs are path dependent

e.g./ MBS – as rates drop, prepayments increase

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LOS h
- describe

1. Simulate numerous paths
   - volatility assumption
   - probability assumption (on how rates will evolve)
2. generate spot rates from the simulated future 1-period interest rates
3. Determine cash flows along each path
4. Calculate PV for each path
5. Calculate avg. PV

- produces market value only by chance
- a constant (drift term) is added to each rate on all paths such that avg. PV = current market value

(i.e. fit to current spot curve and drift adjusted)

## Term Structure Models

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LOS i
- describe

• term structure models provide a description of how interest rates evolve

(use) - valuation/hedging of fixed-income securities and their derivatives

• Interest rate factor: one factor → the short (one-period) rate
  - implies all rates move in the same direction but not necessarily by the same amount

• Interest rate process: stochastic (outcomes involve some randomness)

drift term    stochastic term

$$dr = \underbrace{\theta_t dt}_{\text{expected rate path}} + \sigma_1 dZ$$

expected rate path
- may be constant or mean reverting

randomness (Brownian motion)
- stationary, independent, normally distributed increments

∴ possible to produce negative rates

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LOS i
- describe

• Class of Model/

1/ Arbitrage-free models - begin with observed market prices of a reference set of rates
- assume they are priced correctly
- the model is then calibrated so that it generates the observed reference rates (model price = market price → arb-free)
- does not attempt to explain the observed yield curve, takes it as given

2/ Equilibrium term-structure models - describe term structure dynamics (i.e. change over time) using fundamental economic variables that are assumed to affect interest rates
- parameters are not forced to values that produce current bond prices (model prices consistent with each other only)
- much more parsimonious than arb-free models

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LOS i
- describe

• Equilibrium models: 1/ Cox-Ingersoll-Ross (CIR)

$(\theta - r_t) = 0$ when $\theta = r_t$

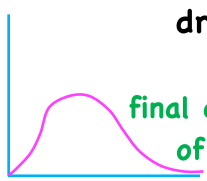
$$dr_t = \underbrace{k(\theta - r_t)}_{\text{mean reverting drift}}dt + \sigma\sqrt{r_t}dZ$$

long run mean rate

rate at time t

modulates speed at which $r_t$ reverts to the mean

random component varies as rates change
- at low rates, the term becomes small
- prevents neg. rates



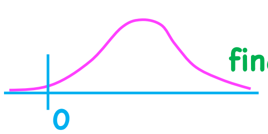
final dist. of $r_t$

2/ Vasicek model

$$dr_t = k(\theta - r_t)dt + \sigma dZ$$

mean reverting

constant vol. over the period



final dist. of $r_t$

• possible for rates to be negative

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LOS i
- describe

• Arbitrage-free models/ 1/ Ho-Lee model

$$dr_t = \theta_t dt + \sigma dZ$$

constant volatility
∴ negative rates are possible

drift term is time dependent
- not constant
- there is a value of $\theta$ for each t - needed to match the current term structure
- not mean reverting

2/ Kalotay-Williams-Fabozzi (KWF)

$$d\ln(r_t) = \underbrace{\theta_t dt + \sigma dZ}_{\text{same as Ho-Lee}}$$

• log of the short rate ($\sim N(\mu, \sigma)$)
- implies short rate is distributed log-normal

∴ prevents negative rates

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LOS i
- describe

| Model | Type | Short Rate | Drift Term | Volatility | neg. rates |
|---|---|---|---|---|---|
| CIR | Equilibrium | $dr_t$ | Mean reversion at speed $k$ | Varies with $\sqrt{r_t}$ | No |
| Vasicek | Equilibrium | $dr_t$ | Mean reversion at speed $k$ | Constant | Yes |
| Ho–Lee | Arbitrage free | $dr_t$ | Time dependent | Constant | Yes |
| KWF | Arbitrage free | $d\ln(r_t)$ | Time dependent | Constant | no |

# Valuation and Analysis of Bonds with Embedded Options

a. describe fixed-income securities with embedded options

b. explain the relationships between the values of a callable or putable bond, the underlying option-free (straight) bond, and the embedded option

c. describe how the arbitrage-free framework can be used to value a bond with embedded options

d. explain how interest rate volatility affects the value of a callable or putable bond

e. explain how changes in the level and shape of the yield curve affect the value of a callable or putable bond

f. calculate the value of a callable or putable bond from an interest rate tree

g. explain the calculation and use of option-adjusted spreads

h. explain how interest rate volatility affects option-adjusted spreads

i. calculate and interpret effective duration of a callable or putable bond

j. compare effective durations of callable, putable, and straight bonds

k. describe the use of one-sided durations and key rate durations to evaluate the interest rate sensitivity of bonds with embedded options

l. compare effective convexities of callable, putable, and straight bonds

m. calculate the value of a capped or floored floating-rate bond

n. describe defining features of a convertible bond

o. calculate and interpret the components of a convertible bond's value

p. describe how a convertible bond is valued in an arbitrage-free framework

q. compare the risk-return characteristics of a convertible bond with the risk-return characteristics of a straight bond and of the underlying common stock

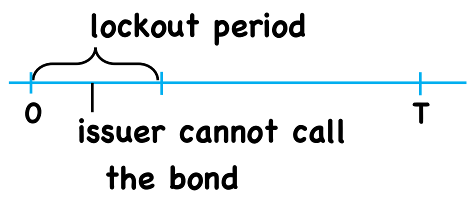
# Embedded Options

Page 1
LOS a
- describe

- contingency provisions that can be exercised by the holder or issuer, or automatically, depending on the course of interest rates

⇒ these options are not independent of the bond and, hence, cannot be traded separately

1) Call options ⇒ callable bonds → benefits issuer




European Call
- single date

American Call
- continuously

Bermuda Call
- multiple dates (predetermined)

e.g./ GSE (Fannie Freddie)
- callable at 100% of par
- lockout periods - 3 mos. → 1 yr.
- Bermuda style

Page 2
LOS a
- describe

2) Put options → putable bond → benefits holder
- usually at par
- typically European, sometimes Bermuda, but not American

→ extendible bond
- sort of like a put ⇒ holder can redeem or extend

3) Complex Options/
⇒ Callable and Putable
⇒ Convertible (also Callable)
⇒ Contingent Options → e.g. estate put
⇒ Sinking Fund Provision (usually also Callable)
+ acceleration provision (retire faster)
+ delivery option (buy back disc. bonds for the fund)

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LOS a
- describe

Investors in putable bonds most likely seek to take advantage of:
A. interest rate movements
B. changes in the issuer's credit rating
C. movements in the price of the issuer's common stock

The decision to exercise the option embedded in an extendible bond is made by:
A. the issuer
B. the bondholder
C. either the issuer or the bondholder

The conversion option is a convertible bond is a right held by:
A. the issuer
B. the bondholders
C. jointly by the issuer and the bondholders

## Relationships

Page 3
LOS b
- explain

| (A) | | (B) | | (C) |
|---|---|---|---|---|
| Value of Callable Bond | = | Value of Straight Bond | − | Value of issuer call option |

∴ C = B − A

arb-free value

| | | | | |
|---|---|---|---|---|
| Value of Putable Bond | = | Value of Straight Bond | + | Value of investor put option |
| (D) | | (E) | | (F) |

∴ F = D − E

Page 4
LOS c
- describe

- Chapter example/

3 yr., 4.25% annual

| Par | Spot | Forward |
|---|---|---|
| 2.5 | 2.5 | 2.5 |
| 3.0 | 3.008 | 3.518 |
| 3.5 | 3.524 | 4.564 |

$$\frac{4.25}{1.025}+\frac{4.25}{(1.03008)^2}+\frac{104.25}{(1.03524)^3}=102.114$$

$$\underbrace{\frac{4.25}{1.025}+\frac{4.25}{(1.025)(1.03518)}+\frac{104.25}{(1.025)(1.03518)(1.04564)}}_{\text{better approach for bond w/options}}=102.114$$

e.g./ - callable each yr. if $V_t > 100$, $\sigma = 0\%$

Value of Call

$$\begin{array}{r} 102.114 \\ \underline{-\,101.707} \\ 0.407 \end{array}$$

end of

| Today | Year 1 | Year 2 |
|---|---|---|
| $\frac{100+4.25}{1.025} = 101.707$ | $\frac{99.70+4.25}{1.03518} = 100.417$ | $\frac{104.25}{1.04564} = 99.70$ |
| | - called at 100 | |

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LOS c
- describe

- Chapter example/

3 yr., 4.25% annual

| Par | Spot | Forward |
|---|---|---|
| 2.5 | 2.5 | 2.5 |
| 3.0 | 3.008 | 3.518 |
| 3.5 | 3.524 | 4.564 |

$$\frac{4.25}{1.025}+\frac{4.25}{(1.03008)^2}+\frac{104.25}{(1.03524)^3}=102.114$$

$$\underbrace{\frac{4.25}{1.025}+\frac{4.25}{(1.025)(1.03518)}+\frac{104.25}{(1.025)(1.03518)(1.04564)}}_{\text{better approach for bond w/options}}=102.114$$

e.g./ - putable each yr. if $V_t < 100$, $\sigma = 0\%$

Value of put.

$$\begin{array}{r} 102.397 \\ \underline{-\,102.114} \\ 0.283 \end{array}$$

end of

| Today | Year 1 | Year 2 |
|---|---|---|
| $\frac{100.707+4.25}{1.025} = 102.397$ | $\frac{100+4.25}{1.03518} = 100.707$ | $\frac{104.25}{1.04564} = 99.70$ |
| | | Put at 100 |

$$\times 1.025 \times 1.03518 = .30$$

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LOS c
- describe

**Bond A – option free**
**Bond B – callable at par (yr 2 and yr 3)**
**Bond C – callable and putable at par (yr 2 and yr 3)**

**Relative to the value of Bond A, the value of Bond B is:**
**A. lower**
**B. the same**
**C. higher**

**Relative to the value of Bond B, the value of Bond C is:**
**A. lower**
**B. the same**
**C. higher**

**Under a steeply upward-sloping yield curve, Bond C will most likely:**
**A. be called by the issuer**
**B. be put by the bondholders**
**C. mature without exercise of any of the embedded options**

## Interest Rate Volatility

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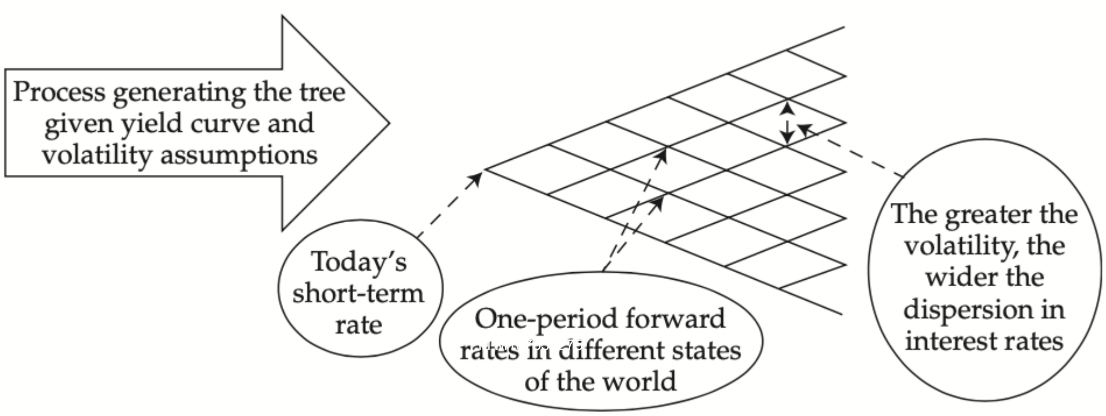
LOS d
- explain

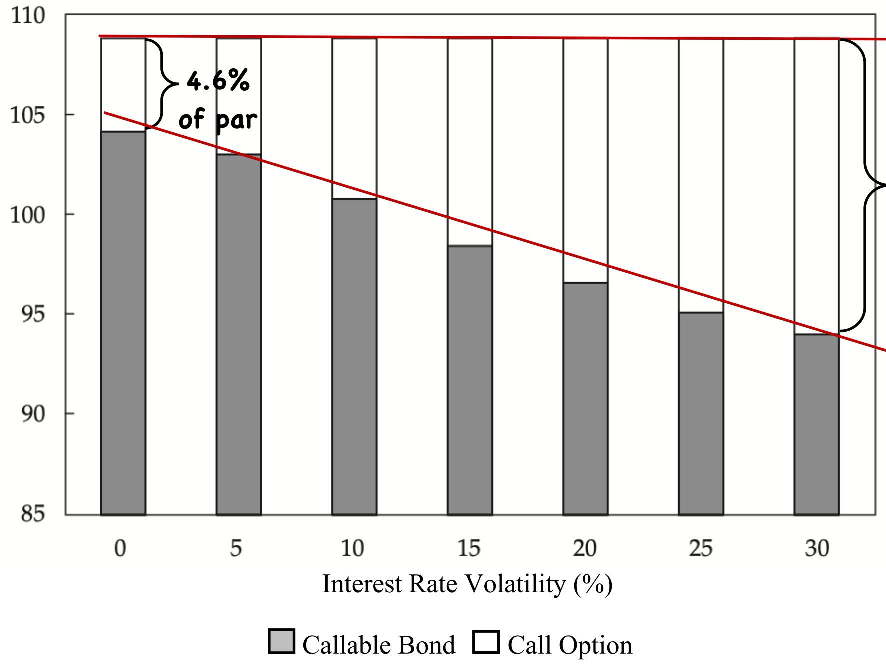
- value of embedded option increases
  with interest rate volatility



Page 8
LOS d
– explain

– 30-yr., 4.5%, Callable at par → 10 yrs.

assume/ 4% flat yield curve




14.78% of par

**Note:** Level Bond Value (straight bond)

• increasing call option value
• decreasing Callable Bond Value

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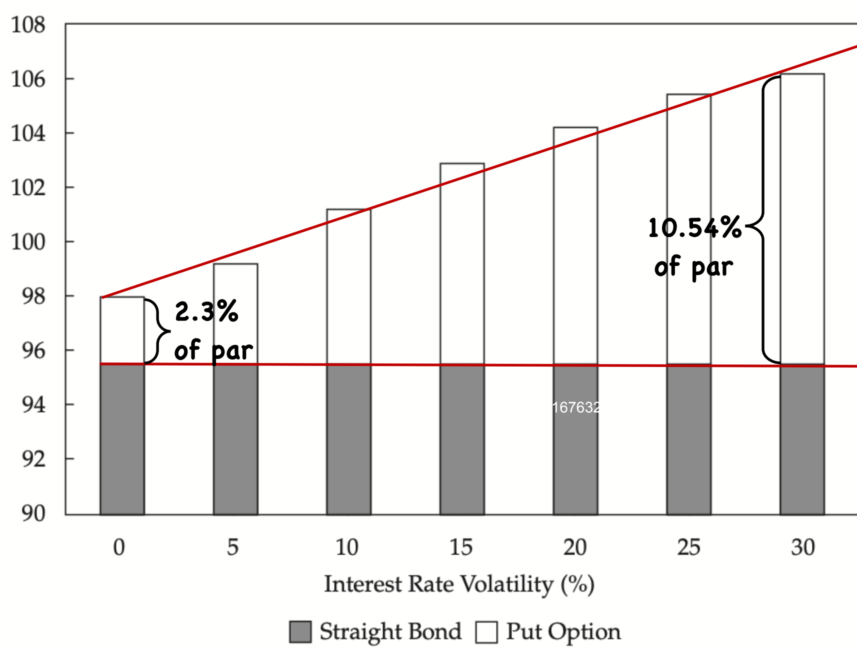
LOS d
– explain

– 30-yr., 3.75%, Putable at Par in 10 years

assume → 4% flat yield curve




• increasing value of put option
• increasing value of Putable Bond

Level Value of Straight Bond
– unaffected by volatility

## Level/Sharpe

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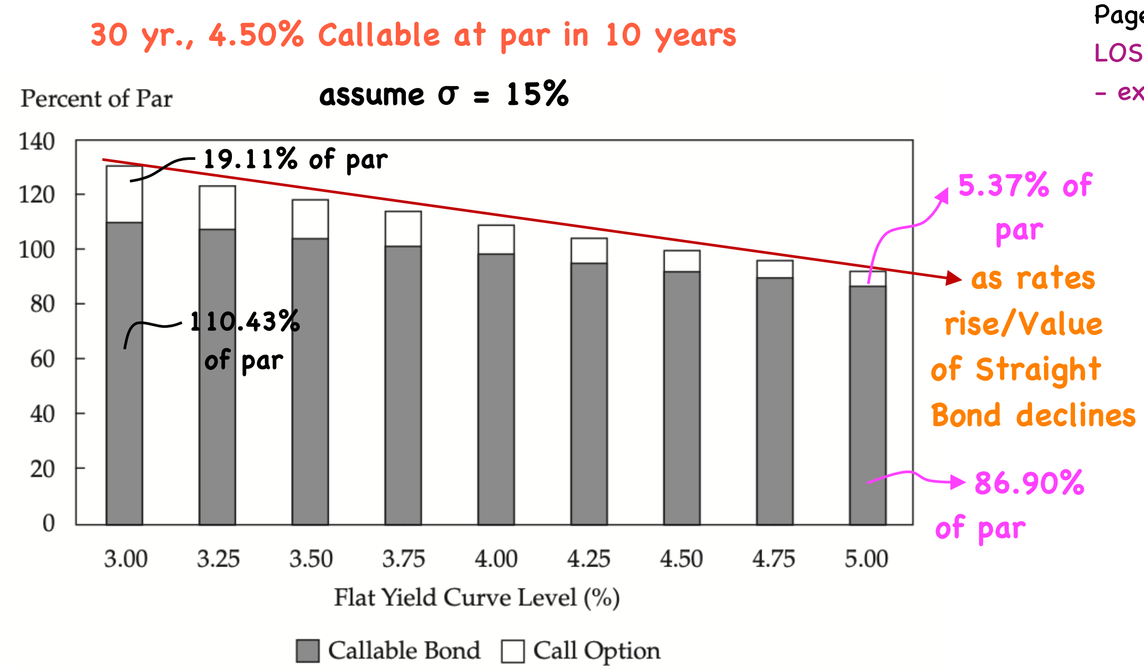
LOS e
- explain

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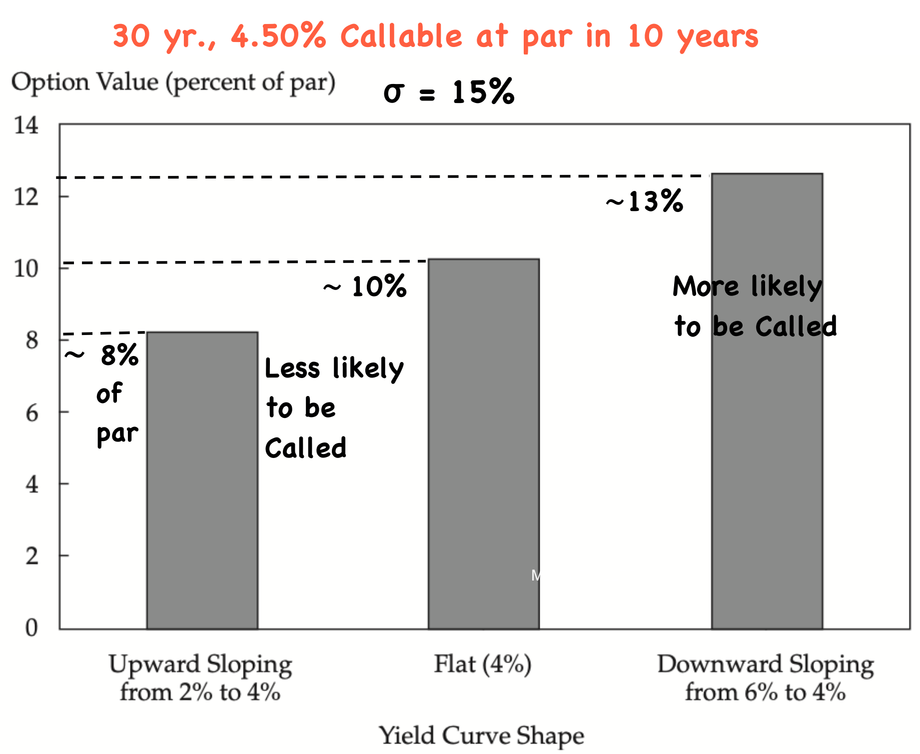
- explain




- assuming normal upward-sloping yield curve, callable bond issued at par = call option OTM

- if σ = 0%, never would be called

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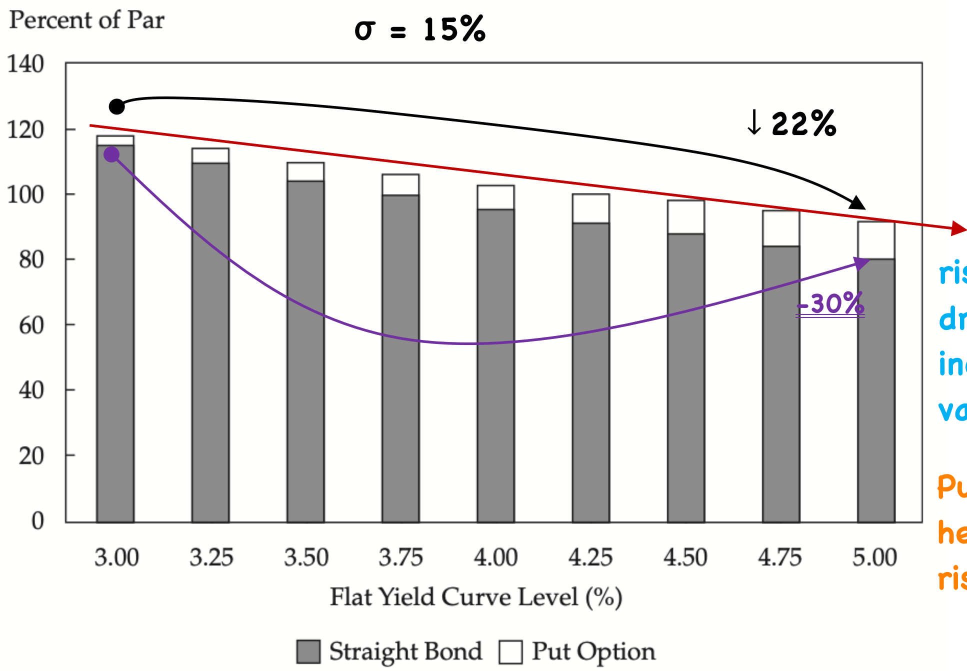
LOS e
- explain

30 yr., 3.75% Putable at par in 10 yrs.




as rates rise, bond values drop, put increases in value

Put option is a hedge against rising rates

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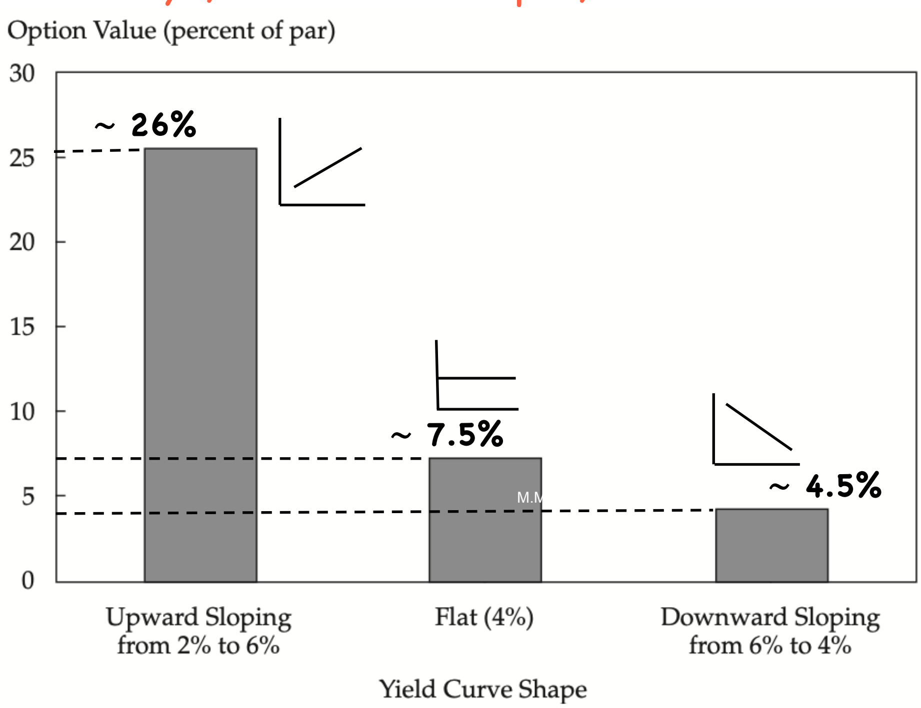
LOS e
- explain

30 yr., 3.75% Putable at par , σ = 15%




- only when rates rise above 3.75% would it potentially make sense to use the put.

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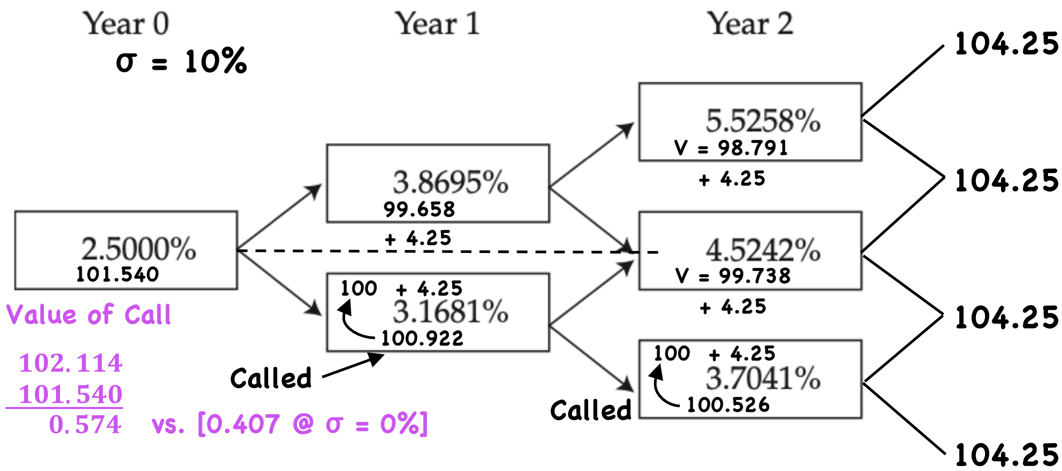
LOS f
- calculate

Par 3 yr., 4.25% Callable at par (yr. 1 & yr. 2)

2.5%
3.0%
3.5%




Page 15a
LOS f
- calculate

Assume that volatility is now 15%.

the new value of the callable bond is:

A. < 101.540
B. = 101.540
C. > 101.540

The bond is now callable at 102.

The new value of the callable bond is:

A. 100.00
B. 102.00
C. 102.114

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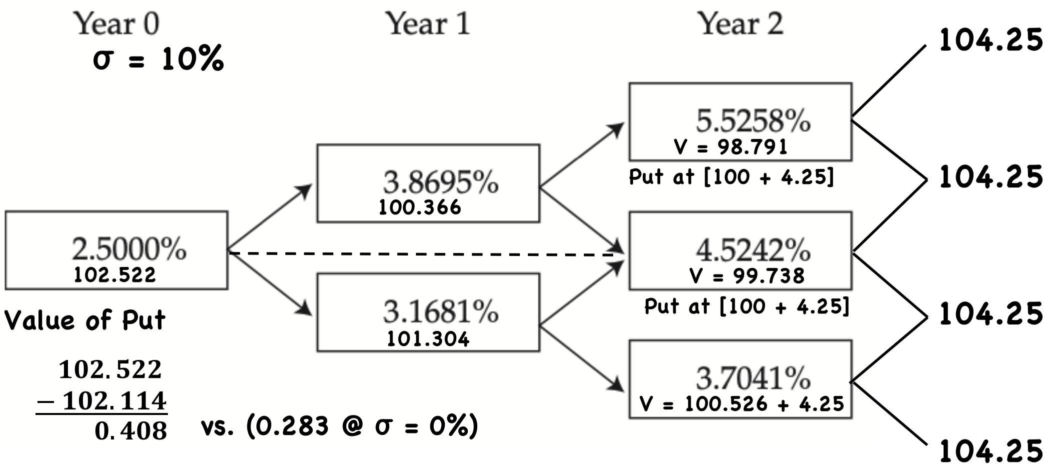
LOS f
- calculate

Par 3 yr., 4.25% Putable at par (yr. 1 & yr. 2)

**2.5%**
**3.0%**
**3.5%**




Value of Put

$$\begin{array}{r} 102.522 \\ \underline{-\,102.114} \\ 0.408 \end{array}$$

vs. (0.283 @ $\sigma$ = 0%)

---

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LOS f
- calculate

**Assume that volatility is now 20%**

**The new value of the putable bond is:**

**A. < 102.522**
**B. = 102.522**
**C. > 102.522**

**The bond is now putable at 95.**

**The value of the bond is closest to:**

**A. 97.522**
**B. 102.114**
**C. 107.522**

# Option-Adjusted Spread

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LOS g
- explain

- extends the valuation framework to the valuation of risky bonds

- raise the one-year forward rates derived from the default-free benchmark yield curve by a $\underbrace{\text{fixed-spread}}_{\text{estimated from the market price of suitable bonds of similar credit quality}}$ (Z-spread)

Recall/

$$\frac{4.25}{1.025}+\frac{4.25}{(1.025)(1.03518)}+\frac{104.25}{(1.025)(1.03518)(1.04564)}=102.114$$

- with a 100 bps Z-spread

$$\frac{4.25}{1.035}+\frac{4.25}{(1.035)(1.04518)}+\frac{104.25}{(1.035)(1.04518)(1.05564)}=99.326$$

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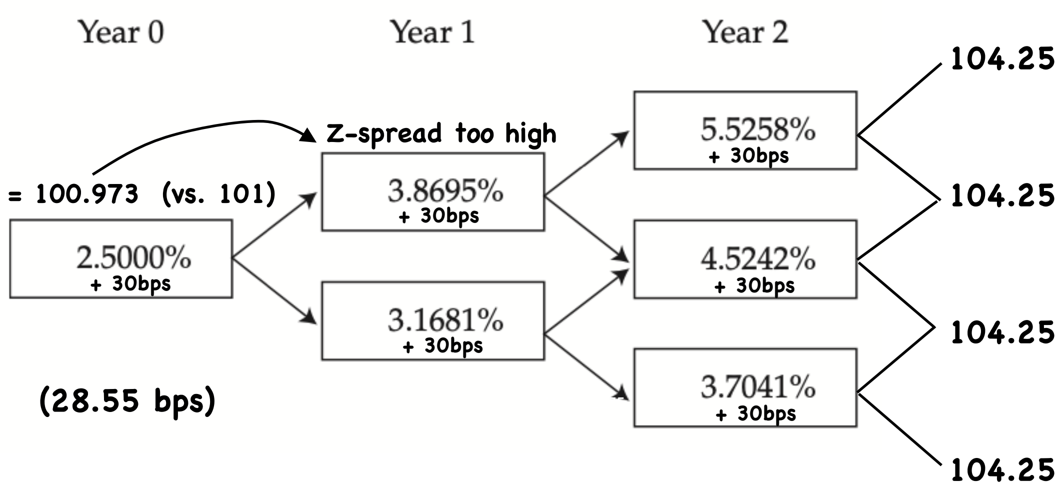
LOS g
- explain

$\underbrace{\text{3-yr., 4.25\% annual, callable at par}}_{\text{now risky}}$ , σ = 10% @ 101.000

• Z-spread = 30bps



# OAS & Volatility

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LOS h
- explain

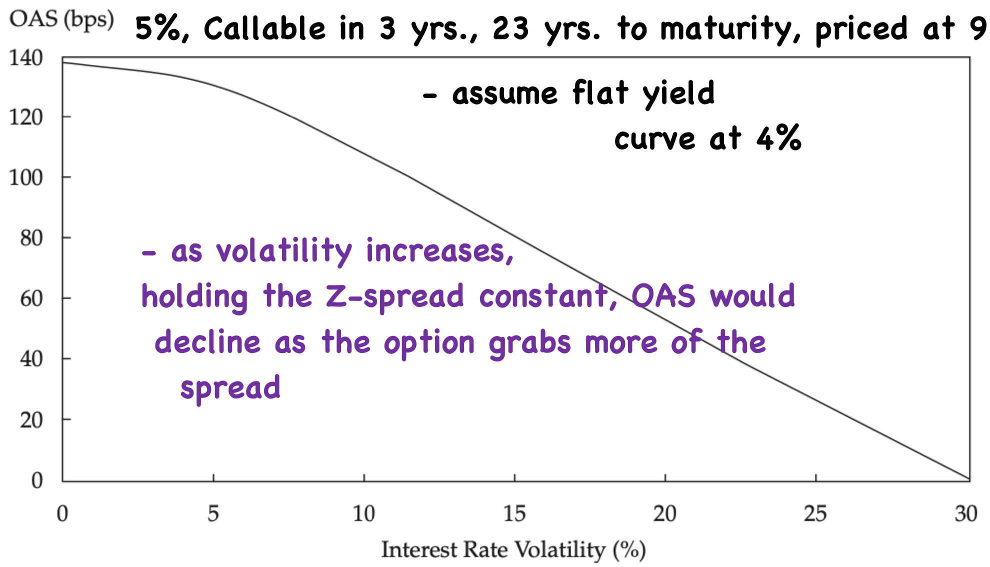
- OAS is volatility dependent

Z-spread = OAS - option cost

5%, Callable in 3 yrs., 23 yrs. to maturity, priced at 95

- assume flat yield curve at 4%

- as volatility increases, holding the Z-spread constant, OAS would decline as the option grabs more of the spread




# Duration

Page 19
LOS i
- calculate
- interpret

⇒ for bonds with embedded options, only appropriate measure is:

$$\text{EffDur} = \frac{(PV_-) - (PV_+)}{2\Delta\text{curve} \cdot P_0}$$

1) given $P_0$, calculate OAS given $\sigma$

2) Shift the curve down, generate a new tree, then revalue the bond using OAS from 1) $PV_-$

3) shift curve up, repeat 2) $PV_+$

e.g./ 3 yr., 4.25% Callable, $\sigma$ = 10%
$P_0$ = 101.00 OAS = 28.55 bps

⇒ $PV_-$ $\Delta$curve = 30bps ↓
= 101.599

⇒ $PV_+$ $\Delta$curve = 30bps ↑
= 100.407

$$\text{EffDur} = \frac{101.599 - 100.407}{2 \times 0.003 \times 101} = 1.97$$

∴ 100bps ↑ rates, ↓ 1.97% in bond price

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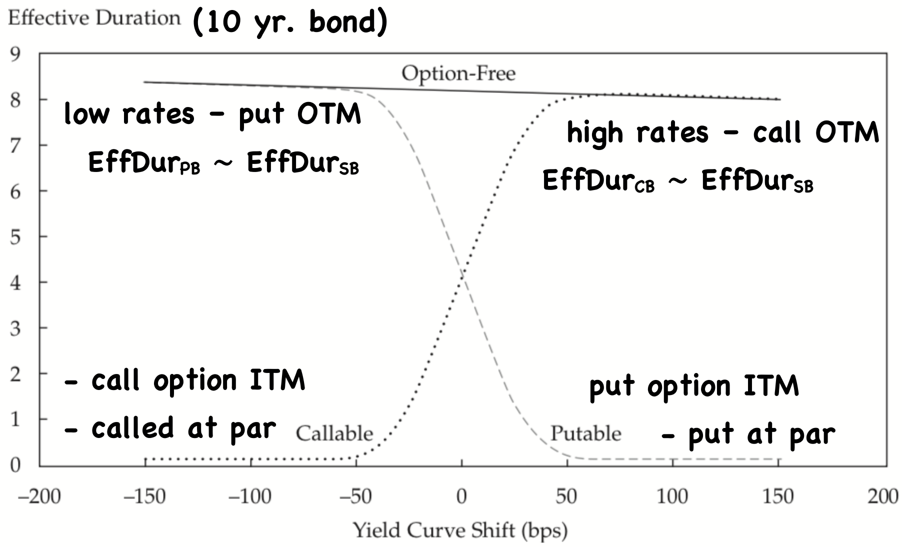
LOS j
- compare

⇒ EffDur of a callable/putable bond $\lesssim$ EffDur of a straight bond




## One-Sided/Key-Rate Duration

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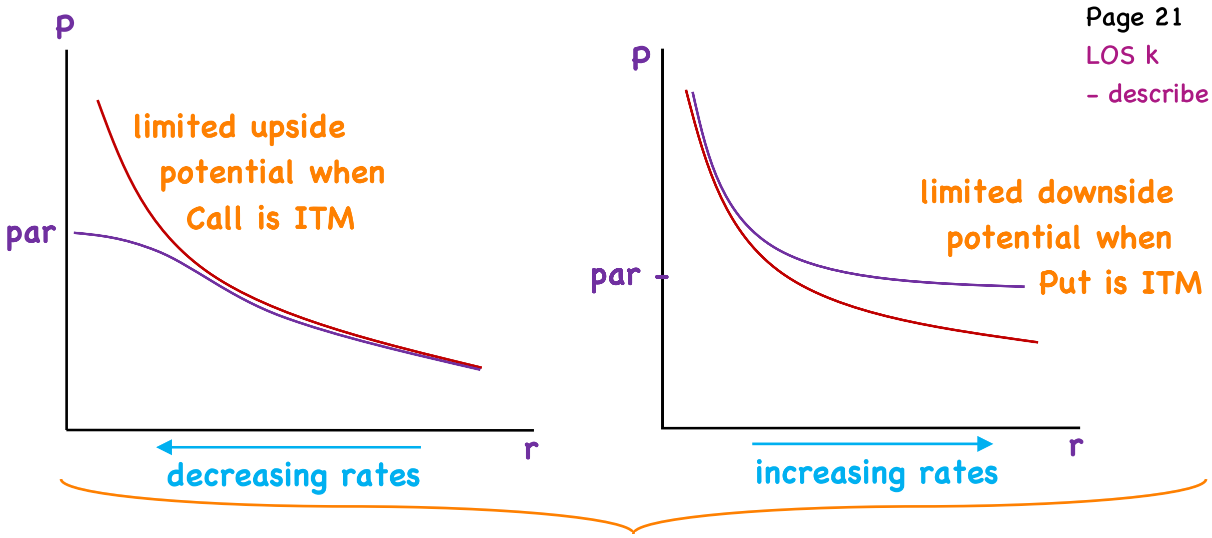
LOS k
- describe




price sensitivity of bonds with embedded options is not symmetrical to equal $^{+}/_{-}$ Δcurve

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LOS k
- describe

e.g./ 5-yr., 4.25% annual, Callable at Par
4% flat curve, σ = 15%

| | | +Δcurve = 30bps | −Δcurve = 30bps |
|---|---|---|---|
| Value of Bond | 99.75 | 99.17 | 100.00 |
| Duration Measure | EffDur | 1S-Dur | 1S-Dur |
| | 1.39 | 1.94 | 0.84 |

more sensitive to interest rate increases

5-yr,. 4.1% annual, Putable at par, 4% flat curve, σ = 15%

| | | +Δcurve = 30bps | −Δcurve = 30bps |
|---|---|---|---|
| Value of Bond | 100.45 | 100 | 101.81 |
| Duration Measure | EffDur | 1S-Dur | 1S-Dur |
| | 3.00 | 1.49 | 4.51 |

more sensitive to interest rate declines

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- describe

• Key Rate/ • EffDur assumes parallel shifts in the benchmark yield curve

- How would bond value change if only the 2-year moved up by 5 bps?

- Key Rate Duration (partial duration)

help identify shaping risk (Steepening, fattening)

$$\sum_{i=1}^{n} \text{KeyRateDur}_i = \text{EffDur}$$

- for a given Δcurve at each key rate

∴ valuation procedure and formula identical to full EffDur

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- describe

| Coupon (%) | Price (% of par) | Key Rate Durations | | | | |
|---|---|---|---|---|---|---|
| | | Total | 2-Year | 3-Year | 5-Year | 10-Year |
| 0 | 67.30 | 9.81 | −0.07 | −0.34 | −0.93 | 11.15 |
| 2 | 83.65 | 8.83 | −0.03 | −0.13 | −0.37 | 9.37 |
| 4 | 100.00 | 8.18 | 0.00 | 0.00 | 0.00 | 8.18 |
| 6 | 116.35 | 7.71 | 0.02 | 0.10 | 0.27 | 7.32 |
| 8 | 132.70 | 7.35 | 0.04 | 0.17 | 0.47 | 6.68 |
| 10 | 149.05 | 7.07 | 0.05 | 0.22 | 0.62 | 6.18 |

10-yr. Option Free Bond
4% flat curve

30 yr. bond, Callable in 10 yrs., $\sigma$ = 15%, 4% flat curve

| Coupon (%) | Price (% of par) | Key Rate Durations | | | | | |
|---|---|---|---|---|---|---|---|
| | | Total | 2-Year | 3-Year | 5-Year | 10-Year | 30-Year |
| 2 | 64.99 | 19.73 | −0.02 | −0.08 | −0.21 | −1.97 | 22.01 |
| 4 | 94.03 | 13.18 | 0.00 | 0.02 | 0.05 | 3.57 | 9.54 |
| 6 | 114.67 | 9.11 | 0.02 | 0.10 | 0.29 | 6.00 | 2.70 |
| 8 | 132.27 | 7.74 | 0.04 | 0.17 | 0.48 | 6.40 | 0.66 |
| 10 | 148.95 | 7.14 | 0.05 | 0.22 | 0.62 | 6.06 | 0.19 |

- not likely to be called

| Year | Par Rates | Spot |
|---|---|---|
| 1 | 4 | 4 |
| 2 | 4 | 4 |
| 3 | 4 | 4 |
| 4 | 4 | 4 |
| 5 → | 5 | 5.107 |
| 6 | 4 | 3.964 |
| 7 | 4 | 3.969 |
| 8 | 4 | 3.973 |
| 9 | 4 | 3.976 |
| 10 | 4 | 3.979 |

r(5)

$$\frac{5}{1.04}+\frac{5}{(1.04)^2}+\frac{5}{(1.04)^3}+\frac{5}{(1.04)^4}$$

$$+\frac{105}{x^5}=100 \qquad x=1.05107466$$

r(6)

$$\frac{4}{1.04}+\frac{4}{(1.04)^2}+\frac{4}{(1.04)^3}+\frac{4}{(1.04)^4}$$

$$+\frac{4}{(1.05107)^5}+\frac{104}{x^6}=100$$

$$x=1.03964278$$

r(7)

$$\frac{4}{1.04}+\frac{4}{(1.04)^2}+\frac{4}{(1.04)^3}+\frac{4}{(1.04)^4}+\frac{4}{(1.05107)^5}+\frac{4}{(1.03964)^6}$$

$$+\frac{104}{x^7}=100 \qquad x=1.0396938 \text{ etc ...}$$

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LOS k
- describe

| Coupon (%) | Price (% of par) | Key Rate Durations | | | | |
|---|---|---|---|---|---|---|
| | | Total | 2-Year | 3-Year | 5-Year | 10-Year |
| 0 | 67.30 | 9.81 | −0.07 | −0.34 | −0.93 | 11.15 |
| 2 | 83.65 | 8.83 | −0.03 | −0.13 | −0.37 | 9.37 |
| 4 | 100.00 | 8.18 | 0.00 | 0.00 | 0.00 | 8.18 |
| 6 | 116.35 | 7.71 | 0.02 | 0.10 | 0.27 | 7.32 |
| 8 | 132.70 | 7.35 | 0.04 | 0.17 | 0.47 | 6.68 |
| 10 | 149.05 | 7.07 | 0.05 | 0.22 | 0.62 | 6.18 |

10-yr. Option Free Bond
4% flat curve

30 yr. bond, Callable in 10 yrs., $\sigma$ = 15%, 4% flat curve

| Coupon (%) | Price (% of par) | Key Rate Durations | | | | | |
|---|---|---|---|---|---|---|---|
| | | Total | 2-Year | 3-Year | 5-Year | 10-Year | 30-Year |
| 2 | 64.99 | 19.73 | −0.02 | −0.08 | −0.21 | −1.97 | 22.01 |
| 4 | 94.03 | 13.18 | 0.00 | 0.02 | 0.05 | 3.57 | 9.54 |
| 6 | 114.67 | 9.11 | 0.02 | 0.10 | 0.29 | 6.00 | 2.70 |
| 8 | 132.27 | 7.74 | 0.04 | 0.17 | 0.48 | 6.40 | 0.66 |
| 10 | 148.95 | 7.14 | 0.05 | 0.22 | 0.62 | 6.06 | 0.19 |

- not likely to be called

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LOS k
- describe

| Coupon (%) | Price (% of par) | Key Rate Durations | | | | |
|---|---|---|---|---|---|---|
| | | Total | 2-Year | 3-Year | 5-Year | 10-Year |
| 0 | 67.30 | 9.81 | −0.07 | −0.34 | −0.93 | 11.15 |
| 2 | 83.65 | 8.83 | −0.03 | −0.13 | −0.37 | 9.37 |
| 4 | 100.00 | 8.18 | 0.00 | 0.00 | 0.00 | 8.18 |
| 6 | 116.35 | 7.71 | 0.02 | 0.10 | 0.27 | 7.32 |
| 8 | 132.70 | 7.35 | 0.04 | 0.17 | 0.47 | 6.68 |
| 10 | 149.05 | 7.07 | 0.05 | 0.22 | 0.62 | 6.18 |

10-yr. Option Free Bond
4% flat curve

30 yr., Putable in 10 yrs., 4% flat curve, $\sigma$ = 15%

| Coupon (%) | Price (% of par) | Key Rate Durations | | | | | |
|---|---|---|---|---|---|---|---|
| | | Total | 2-Year | 3-Year | 5-Year | 10-Year | 30-Year |
| 2 | 83.89 | 9.24 | −0.03 | −0.14 | −0.38 | 8.98 | 0.81 |
| 4 | 105.97 | 12.44 | 0.00 | −0.01 | −0.05 | 4.53 | 7.97 |
| 6 | 136.44 | 14.75 | 0.01 | 0.03 | 0.08 | 2.27 | 12.37 |
| 8 | 169.96 | 14.90 | 0.01 | 0.06 | 0.16 | 2.12 | 12.56 |
| 10 | 204.38 | 14.65 | 0.02 | 0.07 | 0.21 | 2.39 | 11.96 |

likely to be put. Behaves like 10-yr. option-free bond.

# Effective Convexity

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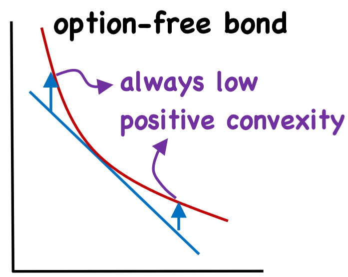
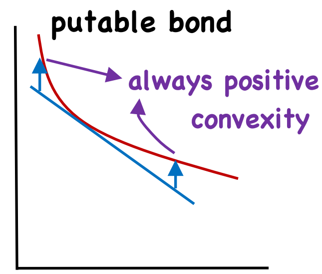
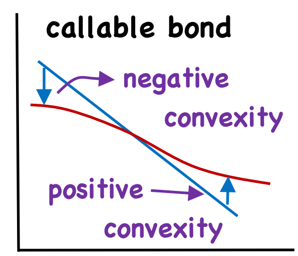
- compare

Recall/ Duration is a linear approx. of price changes

$$\text{EffCon} = \frac{(PV_-) + (PV_+) - 2PV_0}{(\Delta curve)^2 \cdot PV_0}$$




P↑ with −Δcurve >

P↓ with +Δcurve

(floored at put price)

when call in near the money, EffCon < 0
(capped at call price)

---

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LOS L
- compare

3-yr., 3.75% annual

| | $X_{call}$ | $Y_{put}$ |
|---|---|---|
| $PV_0$ | 100.594 | 101.330 |
| $PV_-$ | 101.194 | 101.882 |
| $PV_+$ | 99.86 | 100.925 |

(Δcurve = 30bps) applies to $PV_-$ and $PV_+$

1. EffDur & EffCon of X & Y?

| | X | Y |
|---|---|---|
| $\text{EffDur} = \frac{(PV_-) - (PV_+)}{2\Delta curve \cdot P_0}$ | $\frac{101.194 - 99.86}{2(.003)(100.594)} = 2.21$ | 1.58 |
| $\text{EffCon} = \frac{(PV_-) + (PV_+) - 2PV_0}{(\Delta curve)^2 PV_0}$ | $\frac{101.194 + 99.86 - 2(100.594)}{(.003)^2 \cdot 100.594}$ | 160.093 |

- .134/.0009

= -148.00

X - Callable Y - Putable

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- compare

When interest rates rise, the effective duration of:

A. Bond X shortens
B. Bond Y shortens
C. the underlying option-free (straight) bond corresponding to Bond X lengthens

When the option embedded in Bond Y is in the money, the one-sided durations most likely show that the bond is:

A. more sensitive to a decrease in interest rates
B. more sensitive to an increase in interest rates
C. equally sensitive to a decrease or to an increase in interest rates

X - Callable Y - Putable

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LOS L
- compare

The price of Bond X is affected:

A. only by a shift in the one-year par rate
B. only by a shift in the three-year par rate
C. by all par rate shifts

The effective convexity of Bond X:

A. cannot be negative
B. turns negative when the embedded option is near the money
C. turns negative when the embedded option moves out of the money

X - Callable Y - Putable

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LOS L
- compare

**Which of the following statements is most accurate?**

**A. Bond Y exhibits negative convexity**

**B. For a given decline in the interest rate, Bond X has less upside potential than Bond Y**

**C. The underlying option-free (straight) bond corresponding to Bond Y exhibits negative convexity**

## Capped/Floored Floaters

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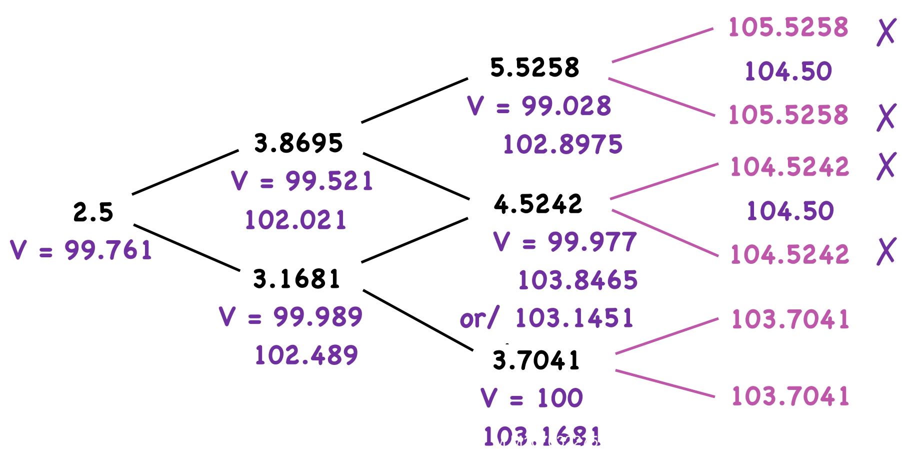
LOS m
- calculate

- cap provision protects the coupon rate from increasing above a specified maximum rate
- protects issuer from rising rates

Value of capped floater = Value of straight bond - Value of embedded cap

e.g./ 4.5% cap, 3-yr. floater, in arrears



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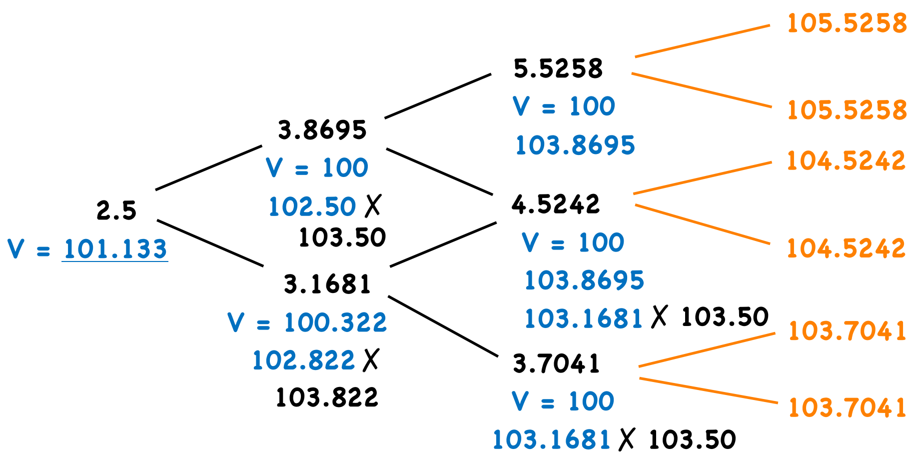
LOS m
- calculate

- floor prevents coupon from decreasing below a specified minimum
- protects investor ⇒ investor option

Value of floored floater = Value of straight bond + Value of embedded floor

e.g./ 3.5% floor, 3-yr. floater, in arrears




## Convertible Bonds

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- describe

⇒ hybrid security

- straight bond + call option on common stock
- holder has the right to convert debt to equity during a 'conversion period' at a 'conversion price'
- issuer
  - can pay a lower coupon
  - if converted, no principal repayment
- if converted ⇒ dilution effects
- if not converted ⇒ issuer must pay back or roll over
  ⇒ bondholders underperform (interest income)

**Excerpt from the Bond's Offering Circular**

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- describe

- **Issue Date:** 3 April 2012 Due: April 3, 2017 5-yr.
- **Status:** Senior unsecured, unsubordinated
- **Interest:** 4.50% of nominal value (par) per annum payable annually in arrears, with the first interest payment date on 3 April 2013 unless prior redeemed or converted
- **Issue Price:** 100% of par denominated into bonds of £100,000 each and integral multiples of £1,000 each thereafter
- **Conversion Period:** $\underbrace{\text{3 May 2012 to 5 March 2017}}_{\text{4 yrs. 10 mos.}}$
- **Initial Conversion Price:** £6.00 per share - conversion price issuance
- **Conversion Ratio:** Each bond of par value of £100,000 is convertible to 16,666.67 ordinary shares
- **Threshold Dividend:** £0.30 per share - only dividends above this amount will affect the conversion price
- **Change of Control Conversion Price:** £4.00 per share ⇒ sometimes its just a 'put at par'
- **Issuer Call Price:** From the second anniversary of issuance (lockout period): 110%; from the third anniversary of issuance: 105%; from the fourth anniversary of issuance: 103% ⇒ Call Option

Another clause:
- hard put
  - cash
- soft put
  - issuer choice
    - cash
    - stock
    - sub. notes

## Components of Value

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LOS o
- calculate
- interpret

⇒ Conversion Value/ (parity value)

$$CV = P_0 \times CR$$

- conversion ratio

$P_0$: share price

e.g./ $CV_1 = 4.58 \times 16{,}666.67 = 76{,}333.33$ (at issuance)

$CV_2 = 6.23 \times 16{,}666.67 = 103{,}833.33$ (Apr. 2013)

⇒ Minimum value of a convertible Bond/ (floor value)

- the greater of
  - conversion value
  - value of the underlying straight bond (found using the arb-free valuation framework)

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- calculate
- interpret

⇒ Market conversion premium/sh./

$$\text{MCP/sh.} = \underbrace{\frac{\overset{\text{bond price}}{PV_0}}{CR}}_{\substack{\text{market} \\ \text{conversion price} \\ \text{(BEP)}}} - \overset{\text{share price}}{P_0}$$

and/ $$\text{MCP ratio} = \frac{\text{MCP/sh.}}{P_0}$$

Market Information

- **Convertible Bond Price on 4 April 2013:** £127,006
- **Share Price on Issue Date:** £4.58
- **Share Price on 4 April 2013:** £6.23
- **Dividend per Share:** £0.16
- **Share Price Volatility per annum as of 4 April 2013:** 25%

e.g./ $$\text{MCP/sh.} = \frac{127,006}{16,666.67} - 6.23$$
$$= 7.62 - 6.23$$
$$= 1.39$$

$$\text{MCP ratio} = \frac{1.39}{6.23}$$
$$= 22.32\%$$

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- calculate
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⇒ Downside risk w/ a Convertible Bond/

$$\text{Premium over straight Value} = \frac{PV_0}{\text{Straight Value}} - 1$$

- assuming straight value = 107,523.95

$$\text{Premium over straight Value} = \frac{127,006}{107,523.95} - 1 = 18.11\%$$

⇒ Upside Potential/

- linear with $P_0$

# Valuation

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LOS p
- describe

Value of Conv. Bond = Value of St. B + call option on issuer's stock

⇒ issuer call (embedded ⇒ force conversion)

Value of Conv. Bond = Value of St. B + stock call option
− issuer call option

⇒ if a put is also embedded

Value of Conv. Bond = Value of St. B + stock call option
− issuer call option
+ holder put option

- Valuation using a binomial tree proceeds the same way but now we need some forecast of share price/period

# Risk-Return

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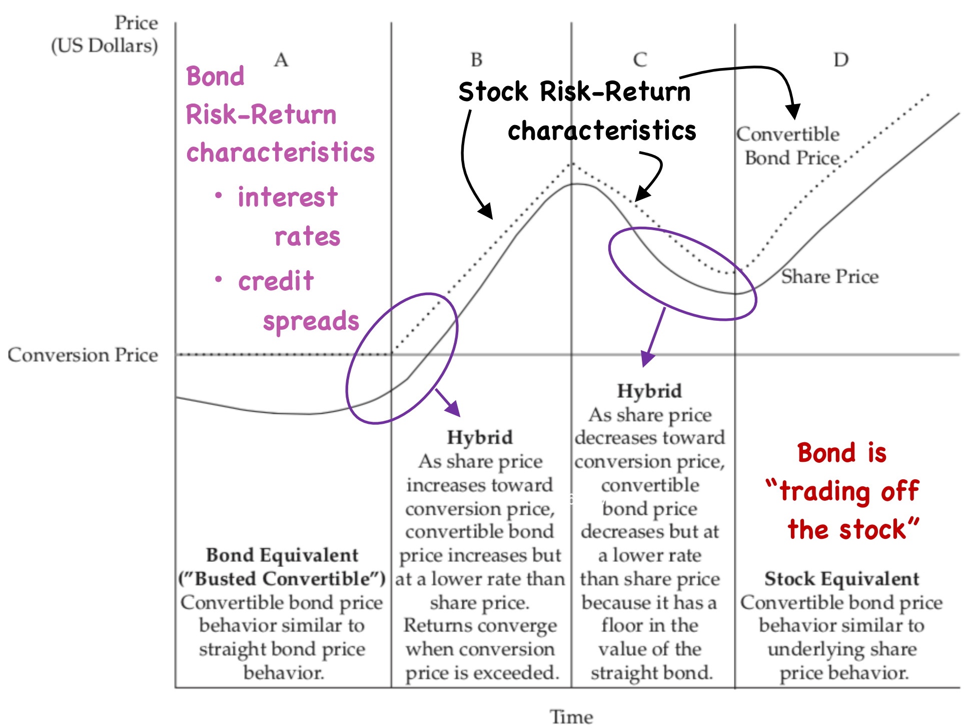
LOS q
- compare



# Credit Analysis Models

a. explain expected exposure, the loss given default, the probability of default, and the credit valuation adjustment

b. explain credit scores and credit ratings

c. calculate the expected return on a bond given transition in its credit rating

d. explain structural and reduced-form models of corporate credit risk, including assumptions, strengths, and weaknesses

e. calculate the value of a bond and its credit spread, given assumptions about the credit risk parameters

f. interpret changes in a credit spread

g. explain the determinants of the term structure of credit spreads and interpret a term structure of credit spreads

h. compare the credit analysis required for securitized debt to the credit analysis of corporate debt

# Credit Analysis Models

Page 1
LOS a
– explain

Corporate bond YTM
benchmark ~~Government~~ bond YTM
} credit spread (G-spread)
– assumes no liquidity or taxation differences

→ default risk – likelihood of an event of default (POD)

→ default risk premium – reflects uncertainty in timing of default

→ credit risk – given default, how much is likely to be lost (LGD)

→ expected exposure – the amount of money that could be lost in default without considering any recovery

e.g./ 1 yr., 4% bond at par → expected exposure = 104

→ recovery rate – %'age recovered in default
– varies by industry, seniority, total leverage, collateral

→ loss severity → (1 – recovery rate)

e.g./ 40% recovery rate, expected exposure = 104 } LGD = 104(1 – .4) = 62.4 per 100 of par value

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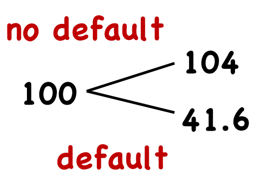
LOS a
– explain




→ probability of default

$$100 = \frac{104(1 - \text{POD}) + 41.6(\text{POD})}{1.03}$$

$1.03 \leftarrow r_f$

$$103 = 104 - 104\text{POD} + 41.6\text{POD}$$

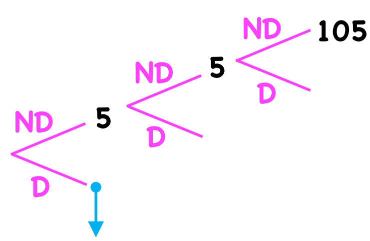
$$\underbrace{\text{POD} = {}^{-1}\!/_{-62.4} = 1.6026\%}_{\text{risk-neutral probability}}$$

vs.

actual probability (1%)

$$\frac{104(.99) + 41.6(.01)}{1.03} = 100.365$$

e.g./ 3 yr., 5% bond @ 104




expected exposure?

$$\underbrace{\left[5 + \frac{5}{1 + r_f} + \frac{105}{(1 + r_f)^2}\right]}\times RR$$

• PV of future CFs
(could have sold it)
• discounted at $r_f$
(risk-neutrality)

• recovery rate → applies to interest as well as principal

Page 3
LOS a
- explain

| Date (1) | Exposure (2) | Recovery (3) | LGD (4) | POD (5) | POS (6) | Expected Loss (7) | DF (8) | PV of Expected Loss (9) |
|---|---|---|---|---|---|---|---|---|
| 0 | | | | | | | | |
| 1 | 88.8487 | 35.5395 | 53.3092 | 1.2500% | 98.7500% | 0.6664 | 0.970874 | 0.6470 |
| 2 | 91.5142 | 36.6057 | 54.9085 | 1.2344% | 97.5156% | 0.6778 | 0.942596 | 0.6389 |
| 3 | 94.2596 | 37.7038 | 56.5558 | 1.2189% | 96.2967% | 0.6894 | 0.915142 | 0.6309 |
| 4 | 97.0874 | 38.8350 | 58.2524 | 1.2037% | 95.0930% | 0.7012 | 0.888487 | 0.6230 |
| 5 | 100.0000 | 40.0000 | 60.0000 | 1.1887% | 93.9043% | 0.7132 | 0.862609 | 0.6152 |
| | | | | 6.0957% | | | CVA = | 3.1549 |

e.g./ 5 yr. zero corporate, $r_f$ = 3.0%, POD = 1.25%

$$T_1 = {}^{100}/_{(1.03)^4} = 88.8487 \times .60 = 53.3092 \times 1.25\% = \frac{0.6664}{1.03} = .6470$$

$$T_2 = {}^{100}/_{(1.03)^3} = 91.5142 \times .60 = 54.9085 \times [(1-1.25\%)1.25\%] = \frac{.6778}{(1.03)^2} = .6389$$

$$FV = {}^{100}/_{(1.03)^5} - CVA = 86.2609 - 3.1549 = 83.1060$$

$$[(1-1.25)^2 1.25\%] = 1.2189\%$$

credit spread = YTM - 3.0% = .7704

where YTM = 3.7704
(N = 5, PMT = 0, FV = 100, PV = -83.1060)

Page 4
LOS a
- explain

| Date (1) | Exposure (2) | Recovery (3) | LGD (4) | POD (5) | POS (6) | Expected Loss (7) | DF (8) | PV of Expected Loss (9) |
|---|---|---|---|---|---|---|---|---|
| 0 | | | | | | | | |
| 1 | 88.8487 | 35.5395 | 53.3092 | 1.2500% | 98.7500% | 0.6664 | 0.970874 | 0.6470 |
| 2 | 91.5142 | 36.6057 | 54.9085 | 1.2344% | 97.5156% | 0.6778 | 0.942596 | 0.6389 |
| 3 | 94.2596 | 37.7038 | 56.5558 | 1.2189% | 96.2967% | 0.6894 | 0.915142 | 0.6309 |
| 4 | 97.0874 | 38.8350 | 58.2524 | 1.2037% | 95.0930% | 0.7012 | 0.888487 | 0.6230 |
| 5 | 100.0000 | 40.0000 | 60.0000 | 1.1887% | 93.9043% | 0.7132 | 0.862609 | 0.6152 |
| | | | | 6.0957% | | | CVA = | 3.1549 |

83.1060 = 35.5395/(1 + IRR) ; IRR = -.5724

$$83.1060 = \underbrace{36.6057/(1 + IRR)^2}_{{}^{0}/_{(1+IRR)} + {}^{36.6057}/_{(1+IRR)^2}} \; ; \; IRR = -.3363$$

83.1060 = $100/(1 + IRR)^5$ ; IRR = 3.77%

annual rates of return depending on when and if default occurs

example #1 & 2

Page 5
LOS b
- explain

→ credit scores - retail lending market - individuals
- small owner operated businesses
- ranks a borrower credit riskiness, but does not provide a POD

→ credit ratings - wholesale lending market - rank credit risk of a company, government, or ABS

3 majors → Moody's Investors Service
→ Standard & Poor's
→ Fitch Ratings

Investment Grade (IG) → 10 ratings
Non-investment Grade (HY) → 10 ratings

issuer and issues
↓
typically for senior unsecured

- issue a letter grade + outlook as well as when the issuer is under 'watch'
- positive
- stable
- negative

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LOS c
- calculate

→ transition matrix

| From/To | AAA | AA | A | BBB | BB | B | CCC,CC,C | D |
|---|---|---|---|---|---|---|---|---|
| AAA | **90.00** | 9.00 | 0.60 | 0.15 | 0.10 | 0.10 | 0.05 | 0.00 |
| AA | 1.50 | **88.00** | 9.50 | 0.75 | 0.15 | 0.05 | 0.03 | 0.02 |
| A | 0.05 | 2.50 | **87.50** | 8.40 | 0.75 | 0.60 | 0.12 | 0.08 |
| BBB | 0.02 | 0.30 | 4.80 | **85.50** | 6.95 | 1.75 | 0.45 | 0.23 |
| BB | 0.01 | 0.06 | 0.30 | 7.75 | **79.50** | 8.75 | 2.38 | 1.25 |
| B | 0.00 | 0.05 | 0.15 | 1.40 | 9.15 | **76.60** | 8.45 | 4.20 |
| CCC,CC,C | 0.00 | 0.01 | 0.12 | 0.87 | 1.65 | 18.50 | **49.25** | 29.60 |
| Credit Spread | 0.60% | 0.90% | 1.10% | 1.50% | 3.40% | 6.50% | 9.50% | |

- can be used to estimate a 1-yr. rate of return given potential credit migration but no default → for 10-year corporate bond

e.g./ assume 10-yr. corporate will have ModDur at horizon end of 7.2 (i.e. - end of YR1)

migrate from A → AA -7.2(.009 - .011) = +1.44%
A → BBB -7.2(.015 - .011) = -2.88%
A → B -7.2(.065 - .011) = -38.88%

.05(A → AAA) + 2.50(A → AA) + 87.50(0) + 8.40(A → BBB) ... + .12(A → C) = -.6342%

∴ E(R) = YTM - .6342% (example #4)

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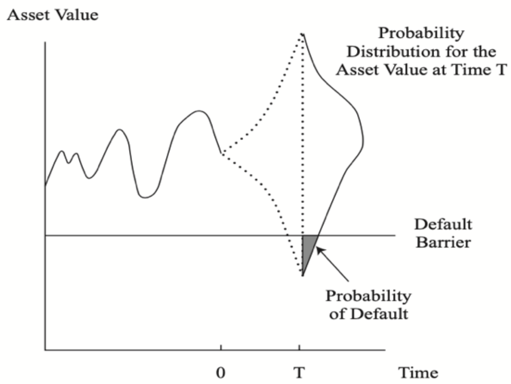
LOS d
- explain

→ Structural models of credit:




- a company defaults on its debt if the value of its assets falls below the amount of its liabilities
- the probability of that event has the features of an option

$A_T = D_T + E_T$

$E_T = \max[A_T - k, 0]$

$D_T = A_T - \max[A_T - k, 0]$

- assumes:
  - company's assets are actively traded (asset value has a lognormal distribution)
  - default depends on the structure of the balance sheet
    i.e. $A_T > D_T$ or $A_T < D_T$ (default is endogenous)

-/ • requires estimates of A and $\sigma_A$ (non-observable)
• assumes D is as reported

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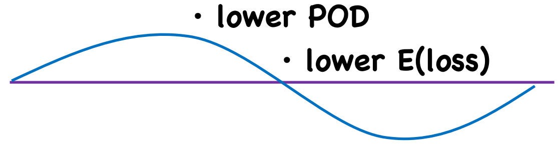
LOS d
- explain

→ Reduced form models - default is on exogeneous variable that occurs randomly

- seek to explain $\underbrace{\text{'when' default occurs}}_{\text{default time-modeled using a Poisson stochastic process}}$, not 'why' (disadv.)

- inputs are observable
- default intensity is estimated using regression analysis on company-specific variables (leverage ratio, NI/assets, cash/assets, etc...) and macroeconomic variables (unemployment, GDP growth rate, ...)
  - default intensity varied depending on company performance and the business cycle
- assumes → at least some of the company's debt trades

business cycle



• lower POD
• lower E(loss)
• higher POD
• higher E(loss)

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LOS e
– calculate

→ valuing risky bonds in an arbitrage-free framework

– introduce volatility and solve for the CVA and credit spread using a binomial interest rate tree

| Maturity | Coupon Rate | Price | Discount Factor | Spot Rate | Forward Rate |
|---|---|---|---|---|---|
| 1 | –0.25% | 100 | 1.002506 | –0.2500% | |
| 2 | 0.75% | 100 | 0.985093 | 0.7538% | 1.7677% |
| 3 | 1.50% | 100 | 0.955848 | 1.5166% | 3.0596% |
| 4 | 2.25% | 100 | 0.913225 | 2.2953% | 4.6674% |
| 5 | 2.75% | 100 | 0.870016 | 2.8240% | 4.9664% |

$$\left(\frac{1.002506}{.985093}\right) - 1 = .017677$$

$$\left(\frac{.985093}{.955848}\right) - 1 = .03596$$

$$\left(\frac{.913225}{.870016}\right) - 1 = .049664$$

$100 = (100 - .25) \times DF_1$

$DF_1 = 100/99.75 = 1.002506$

$100 = (.75 \times DF_1) + (100 + .75) \times DF_2$

$DF_2 = 99.248458/100.75 = .985093$

⋮

$100 = (2.75 \times DF_1) + (2.75 \times DF_2) + (2.75 \times DF_3) + (2.75 \times DF_4) + (100 + 2.75) \times DF_5$

$$\left(\frac{1}{1.002506}\right) - 1 = -.0025$$

$$\left(\frac{1}{.985093}\right)^{1/2} - 1 = .007538$$

$$\left(\frac{1}{.955848}\right)^{1/3} - 1 = .015166$$

$$\left(\frac{1}{.870016}\right)^{1/5} - 1 = .028240$$

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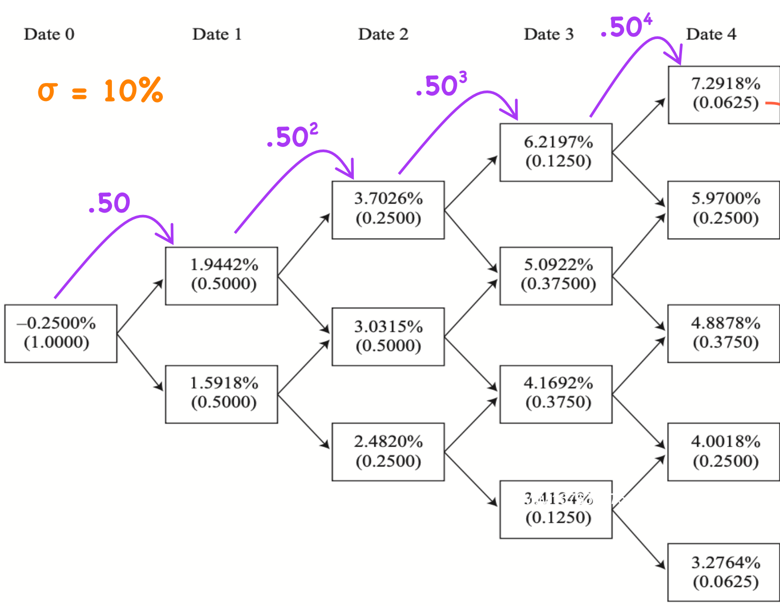
LOS e
– calculate

σ = 10%




probability of obtaining this node

(.0625) × 4

(.0625) × 6

(uuud
uudu
uduu
duuu)

→ 2.75% annual pay 5 yr. gov't. bond (Exhibit #11)

→ 3.5%, annual pay 5 yr. corporate bond (Exhibit #12)

(assuming no default)

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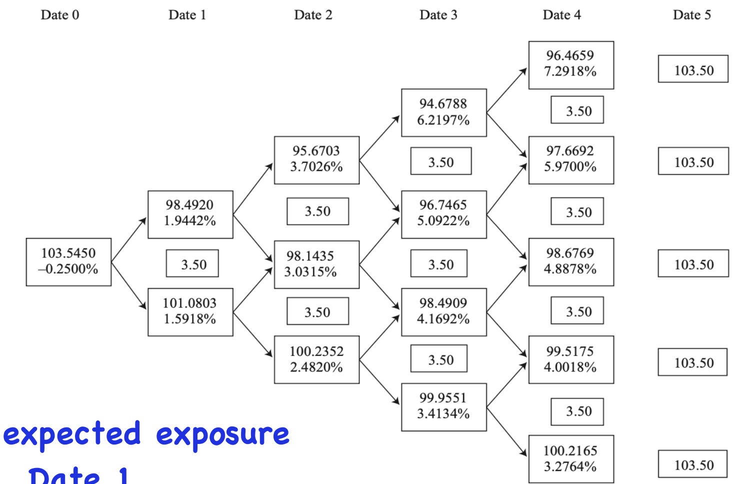
LOS e
– calculate

– calculate the CVA




Date 4:
(.0625 × 96.4659)
+ (.25 × 97.6692)
+ (.375 × 98.6769)
+ (.25 × 99.5175)
+ (.0625 × 100.2165)
+ 3.50
= 102.0931

expected exposure
Date 1
(.5 × 98.4920)
+ (.5 × 101.0803) + 3.50
= 103.2862

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LOS e
– calculate

| Date | Expected Exposure | LGD | POD | Discount Factor | CVA per Year |
|---|---|---|---|---|---|
| 0 | | | given | | |
| 1 | 103.2862 | 61.9717 | 1.2500% | 1.002506 | 0.7766 |
| 2 | 101.5481 | 60.9289 | 1.2344% | 0.985093 | 0.7409 |
| 3 | 101.0433 | 60.6260 | 1.2189% | 0.955848 | 0.7064 |
| 4 | 102.0931 | 61.2559 | 1.2037% | 0.913225 | 0.6734 |
| 5 | 103.5000 | 62.1000 | 1.1887% | 0.870016 | 0.6422 |
| | | × 60% | 6.0957% | CVA = | 3.5394 |

1.25% × (100% – 1.25%) = 1.2344%

1.25% × (100% – 1.25%)$^4$ = 1.1887%

(recovery rate = 40%)

3.50
3.50
60.6260
3.50
(60.9289 × .012344) × .985093
= .740894
61.9717

.7766 ← (61.9717 × .0125) × 1.002506

FV(ND) = 103.5450
CVA – 3.5394
FV(D) 100.0056

N = 5, FV = 100, PMT = 3.5
PV = –100.0056
CPT I/Y = 3.4988%

Credit Spread
3.4988% – 2.75% = .7488%

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LOS e
- calculate

→ summation

- par → spot → forward
- construct interest rate tree using forward rates
- PV(ND) ← calculate PV of corporate bond assuming no default
- calculate expected exposures (× [1 - recovery rate])

= LGD (× POD × $DF_n$)

$-\sum CVA$ ← = CVA/yr.

$= PV(D)$ → find $^I/_Y$ (N = , PMT = , FV = 100, PV = -PV(D))

- $^I/_Y$ - maturity matching gov't. YTM

$\underbrace{\text{= credit spread}}_{\text{assumes spread is based entirely on credit risk}}$

Note: increased volatility has an insignificantly small effect on the CVA

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LOS e
- calculate

- floating rate note → 5 yr., Index + .5% (QM)
→ exhibit 17

| Date | Expected Exposure | LGD | POD | Discount Factor | CVA per Year |
|---|---|---|---|---|---|
| 0 | | | | | |
| 1 | 102.1074 | 81.6859 | 0.5000% | 1.002506 | 0.4095 |
| 2 | 103.6583 | 82.9266 | 0.4975% | 0.985093 | 0.4064 |
| 3 | 104.4947 | 83.5957 | 0.4950% | 0.955848 | 0.3955 |
| 4 | ② 105.6535 | 95.0881 | 0.7388% | 0.913225 | 0.6416 |
| 5 | 105.4864 | 94.9377 | 0.7333% | 0.870016 | 0.6057 |
| | | | 2.9646% | CVA = | 2.4586 |

Changes
YR1 - YR3 = 80%
YR4 - YR5 = 90%

YR1 - YR3 = .5%
YR4 - YR5 = .75%

①
(.0625 × 107.7918) + (.25 × 106.47)
+ (.375 × 105.3878) + (.25 × 104.5018)
+ (.0625 × 103.7764) = 105.4864

② (.0625 × 100.4660)
+ (.25 × 100.4718)
+ (.375 × 100.4767)
+ (.25 × 100.4808)
+ (.0625) × 100.4841)

+ (.125 × 6.7197)
+ (.375 × 5.5922)
+ (.375 × 4.6692)
+ (.125 × 3.9134)

= 105.6535

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LOS e
- calculate

- floating rate note → 5 yr., Index + .5% (QM)
→ exhibit 17

| Date | Expected Exposure | LGD | POD | Discount Factor | CVA per Year |
|---|---|---|---|---|---|
| 0 | | | | | |
| 1 | 102.1074 | 81.6859 | 0.5000% | 1.002506 | 0.4095 |
| 2 | 103.6583 | 82.9266 | 0.4975% | 0.985093 | 0.4064 |
| 3 | 104.4947 | 83.5957 | 0.4950% | 0.955848 | 0.3955 |
| 4 | 105.6535 | 95.0881 | 0.7388% | 0.913225 | 0.6416 |
| 5 | 105.4864 | 94.9377 | 0.7333% | 0.870016 | 0.6057 |
| | | | 2.9646% | CVA = | 2.4586 |

$PV(ND) = 102.3633$
$\sum CVA \quad - 2.4586$
$PV(D) \quad 99.9047$

∴ DM > QM
(discount bond)
(exhibit #19)

Changes
YR1 - YR3 = 80%
YR4 - YR5 = 90%

YR1 - YR3 = .5%
YR4 - YR5 = .75%

.995 / .005 → .50
.995 / .005 → $.995 \times .005 = \underline{.4975\%}$
.995 / .005 → $.995^2 \times .005 = \underline{.4950\%}$
.9925 / .0075 → $.995^3 \times .0075 = \underline{.7388\%}$
.9925 / .0075 → $.995^3 \times .9925 \times .0075 = \underline{.7333\%}$

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LOS f
- interpret

Spread Over the Benchmark Yield:
- Expected Loss from Default
- Liquidity
- Taxation
- Risk Aversion: Compensation for Uncertainty Regarding Expected Loss from Default

microeconomic factors that pertain to the issuer and specific issue

Benchmark Yield:
- Expected Inflation Rate
- Expected Real Rate of Return
- Risk Aversion: Compensation for Uncertainty Regarding Expected Inflation

macroeconomic factors that affect all debt securities

- securities in which it is more difficult to assess POD and recovery rate become less liquid

- as LGD & POD increase, credit spread increases
- as LGD & POD decrease, credit spread decreases

(example #8)

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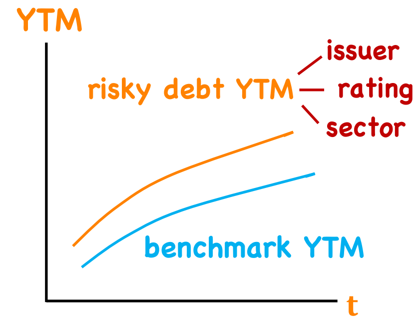
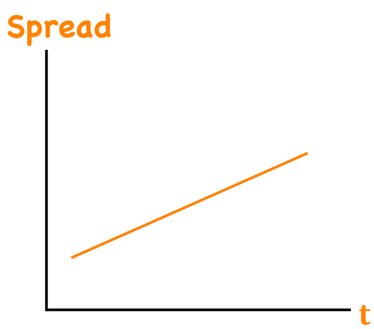
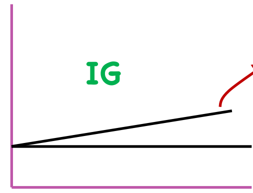
LOS g
- explain



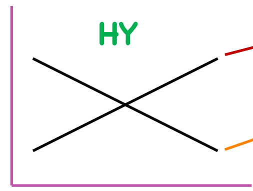

Term Structure of Credit Spreads

1. Credit Quality:



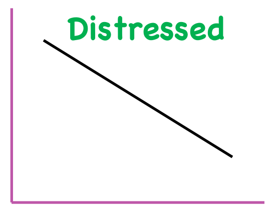

more likely upward sloping for IG

→ stable expectation of default over time




expectation of worsening future economic environment

expectation of better future economic environment




- 100% POD, spread is more of a residual effect of the security trading for its recovery value

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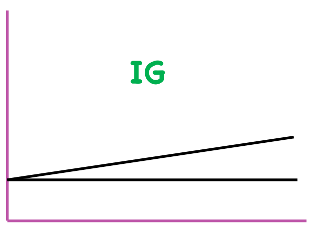
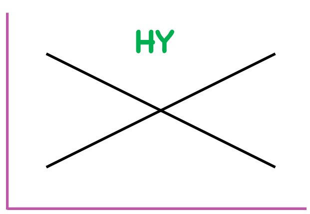
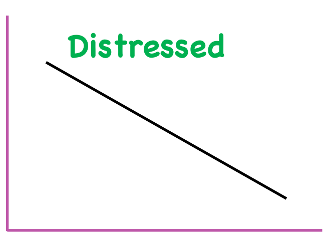
LOS g
- explain




2. Financial Conditions - credit risk of a bond is influenced by expectations for economic growth and inflation

- stronger economy → higher benchmark yields but lower credit spreads for issuers whose POD declines during periods of economic growth

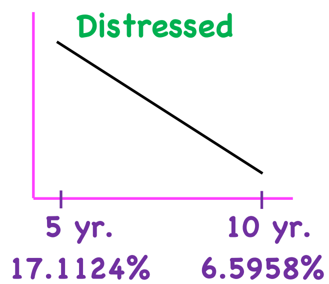
3. Market demand & supply dynamics - credit curve is most heavily influenced by the most frequently traded securities

- infrequently traded → wide bid-ask spread
- frequently traded → narrow bid-ask spread

(both) will distort implied credit spread

4. Company specific risk

Page 19
LOS g
- explain




- just a residual
- no information

e.g./ 5-yr. & 10-yr. zero with POD = 100%

- recovery rate = 40%
- assume a flat benchmark curve at 3.0%

- both will trade for $40 per 100 of par

N = 5 CPT $^{I}/_{Y}$ = 20.1124 if N = 10

PMT = 0 $^{I}/_{Y}$ = 9.5958

FV = 100 spread = 20.1124 spread = 9.5958

PV = -40 $\underline{-\ 3.0000}$ $\underline{-\ 3.0000}$

= 17.1124% 6.5958%

→ Benchmark rates → gov't. spot curve
→ swap curve based on interbank rates often used (greater swap market liquidity for off-the-run securities)

Page 20
LOS h
- compare

→ homogeneity - the degree to which the underlying debt characteristics are similar across individual obligations
- homogenous → general conclusions from the class
- heterogeneous → scrutiny on a loan-by-loan basis

→ granularity - the actual number of obligations in the structured security
- many → draw conclusions based on summary statistics
- fewer → analysis of each individual obligation

→ origination & servicing - exposure to operational & counterparty risk over the life of the securitized asset
(ability of servicer to effectively manage and service the portfolio over the life of the transaction)

→ structure of the secured debt transaction - the SPE + any structural enhancements (e.g. overcollateralization, credit tranching)

# Credit Default Swaps

a. describe credit default swaps (CDS), single-name and index CDS, and the parameters that define a given CDS product

b. describe credit events and settlement protocols with respect to CDS

c. explain the principles underlying and factors that influence the market's pricing of CDS

d. describe the use of CDS to manage credit exposures and to express views regarding changes in the shape and/or level of the credit curve

e. describe the use of CDS to take advantage of valuation disparities among separate markets, such as bonds, loans, equities, and equity-linked instruments

# Credit Default Swaps

Page 1
LOS a
- describe

⇒ credit derivative → underlying = borrower's credit quality

Long ⇒ improving credit quality (Seller of CDS)

Short ⇒ deteriorating credit quality (Buyer of CDS)

- CDS provides protection against default, but also protects against changes in credit quality long before a default

e.g./ credit downgrade

| | credit downgrade |
|---|---|
| Long a bond | - bond loses value |
| Buy a CDS | - CDS gains value |

- Value of a CDS ↑↓ with changing likelihood of default
  ⇒ Actual default may never occur

Page 2
LOS a
- describe

⇒ Definition/ a derivative contract between 2 parties
1) credit protection buyer
2) credit protection seller

in which the CP buyer makes a series of payments (premiums) to the CP seller and receives a promise of compensation for credit losses resulting from default (pre-defined credit event) of a third party

determined and fixed at contract inception

(until default only) → premium leg

Protection Buyer → periodic payments → Protection Seller

Protection Buyer ← payment if default ← Protection Seller

protection leg

Page 3
LOS a
- describe

- Single-name CDS/ a CDS on a specific $\underbrace{\text{borrower}}_{\text{reference entity}}$

• contract specifies a $\underbrace{\text{reference obligation}}$
- a specific debt instrument
- usually senior unsecured
(senior CDS)

⇒ any other debt of equal priority of claims or higher relative to the reference obligation is covered

- payoff of a CDS is determined by the Cheapest-to-deliver obligation
(same seniority as the obligation)

Page 4
LOS a
- describe

Assume that a company with several debt issues trading in the market files for bankruptcy. (i.e. a credit event takes place). What is the CTD obligation for a senior CDS contract?

A. A subordinated unsecured bond trading at 20% of par
B. A 5-yr senior unsecured bond trading at 50% of par
C. A 2-yr senior unsecured bond trading at 45% of par

Page 5
LOS a
- describe

⇒ Index CDS/ • a combination of borrowers

- introduces $\underbrace{\text{'credit correlation'}}_{}$

defaults by certain companies may be connected to defaults by other companies

- the more correlated the defaults, the more costly it is to purchase protection

---

- CDS contracts conform to ISDA specifications

- each contract specifies a 'notional amount'

⇒ total amount of CDS notional can exceed the amount of debt outstanding

⇒ CDS buyer need not be a debtholder

---

Page 6
LOS a
- describe

- CDS has an expiration date

⇒ 1-10 yrs. (5 yrs. most common)

- buyer pays a periodic premium to seller

CDS spread ⇒ a return over Libor
(a.k.a. Credit Spread)

- CDS rates are now standardized, most common are 1% & 5%

1% ⇒ investment grade company/index
5% ⇒ high-yield company/index

- to correct for the standardized rate being too high or too low, an upfront payment/premium is required

PV of difference between the credit spread and the standard rate

Page 7
LOS a
- describe

- spread should be higher
upfront payment

Buyer $\xrightarrow{1\%}$ Seller

upfront payment
- spread should be lower

value of CDS to buyer increases if reference entity's credit quality drops

value of CDS to seller increases if reference entity's credit quality improves

## Credit Events

Page 8
LOS b
- describe

⇒ 3 general types of credit events/

1) bankruptcy - filing
- may not lead to liquidation

2) failure to pay - scheduled interest or principal on any outstanding obligation

3) restructuring - change in seniority, change in currency of payment, reduction of principal/coupon

⇒ must be involuntary (forced on borrower)
- not considered a credit event in the U.S.

- 15 member group within ISDA determines if a credit event occurs (Determinations Committee)

$\left.\begin{array}{l}\text{10 CDS dealer banks}\\ \text{5 end users}\end{array}\right\}$ need 12 votes

Page 9
LOS b
- describe

⇒ Succession event/ • a change in the corporate structure of the reference entity (merger, divestiture, spinoff) in which responsibility for the debt in question becomes unclear

- if an event occurs, CDS contract is modified to reflect the new obligor for the original debt (may even be split among multiple entities)

⇒ Settlement Protocols/

- once an event is declared, both parties have the right, but not an obligation, to settle

- settlement ~ 30 days after the event

Page 10
LOS b
- describe

- Physical settlement
  - holder sells bond to CDS seller at par

- Cash settlement - Seller pays holder for losses

Payout ratio = 1 − Recovery Rate

Payment amount = Payout ratio × Notional

• recovery may occur much later than the payoff to the CDS

⇒ Auction - major banks and dealers submit bids/offers for the CTD defaulted debt
  - identifies the market's expectations for the recovery rate

- CDS parties agree to accept outcome of auction
  - actual recovery may be quite different

Page 11
LOS b
- describe

e.g./ Bankruptcy

- Senior Bonds
  - A trades at 30% of par
  - B trades at 40% of par

- Investor
  - X owns \$10M of Bond A & \$10M of CDS protection
  - Y owns \$10M of Bond B & \$10M of CDS protection

Recovery Rate ⇒ CTD = Bond A = 30% (for both bonds)

Investor X ⇒ receives \$7M, sells or holds the bond for \$3M
(no preference as to settlement)

Investor Y ⇒ receives \$7M, sells or holds the bond for \$4M
(prefers cash settlement)

credit event auction – sets a price for all that choose to cash settle
(2005)

- buyers and sellers must submit an adherence letter to ISDA selecting this option

- requests for physical settlement

- dealers place orders for the debt, Recovery is set at the $\underbrace{\text{market mid-point}}_{\text{price for all those choosing cash settlement}}$

Page 12
LOS b
- describe

⇒ CDS Index Products/

- classified $\underbrace{\text{by region}}$ & $\underbrace{\text{credit quality}}$

125 entities each
CDX - North America - IG (CDX IG)
iTraxx - Europe - Main (iTraxx Main)

$\underbrace{\text{- IG / - Main}}_{\text{investment grade}}$
- quoted in spreads

(100) CDX HY
(50) iTraxx Crossover
high yield - quoted in prices

⇒ Both use standardized coupons and are equally weighted
- settlement is thus $^1/_{125}$ of the notional

- when an entity defaults, it is removed from the index and settled as a single-name CDS

- index then moves forward with a smaller notional

Page 13
LOS b
- describe

⇒ CDS Index Products/

- typically used to protect against/take positions on the credit risk of sectors (bond portfolios that are similar to the index)

e.g./ Sell \$500M of protection on CDX IG
Buy \$3M of protection on component A (single-name CDS)

A defaults/

Investor is: Long $^1/_{125} \times$ 500M = \$4M
Short \$3M
hedged 75% exposure to A

Net exposure ⇒ long \$1M (pays the buyer)

Remaining notional on CDX IG = \$496M

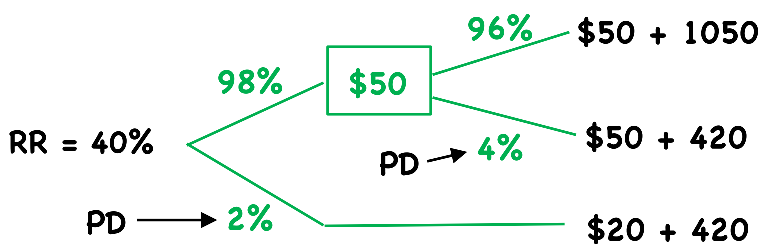
# Valuation & Pricing

Page 14
LOS c
- explain

Recall/ • underlying of a CDS is 'credit' which does not trade
• exists implicitly within the bond

- Hazard rate - probability that an event will occur given that it has not already occurred

2 yr., annual 5% bond/




P = .98 × .96 = 94.08%
(Prob. of Survival)

P = .98 × .04 = 3.92%

P = 2%

Σ = 100%

PD = 5.92%

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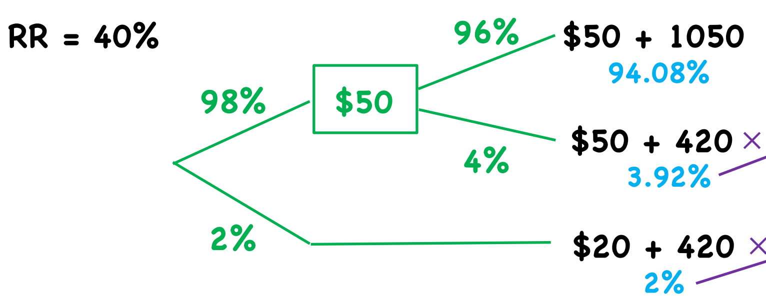
Page 15
LOS c
- explain




LGD
N/A
630
660

expected loss
24.696
13.20
$37.896

PD × LGD = E(loss)

- assume a 10-yr. with equal hazard rate of 2%/yr.

$$PD = 1 - (.98)^{10}$$
$$= 1 - .817$$
$$= 18.3\%$$

Page 16
LOS c
- explain

Assume that a company's hazard rate is a constant 8% per year, or 2% per quarter. An investor sells 5-yr CDS protection on the company with the premiums paid quarterly over the next 5 years.

1. What is the probability of survival for the first quarter?

2. What is the conditional probability of survival for the second quarter?

3. What is the probability of survival through the second quarter?

Page 17
LOS c
- explain

- two legs to a CDS
    1) protection leg – seller pays buyer
    2) premium leg – buyer pays seller (premium)

• CDS spread is calculated by equating:

PV(prot. leg) = PV(pr. leg)

- but since we use standardized rates

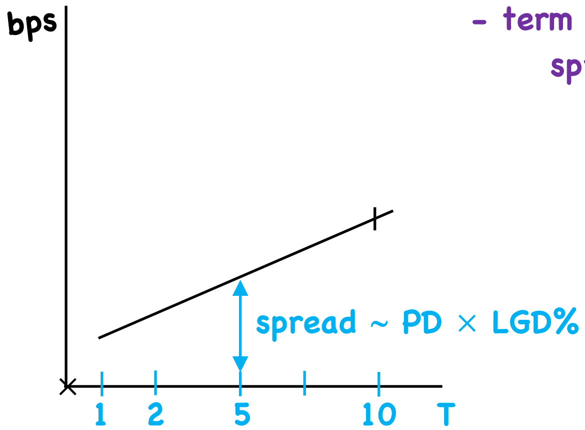
Upfront payment = PV(prot. leg) – PV(pr. leg)

if > 0, buyer pays seller
   < 0, seller pays buyer

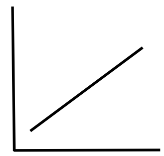
* CFA does not require the calculation of any of the PVs

Page 18
LOS c
- explain

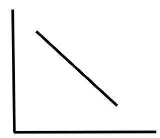
Recall/



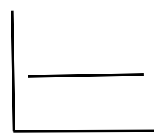

- term structure of credit spreads
  - if for a specific company, called 'credit curve'



greater likelihood of default in later yrs.



greater likelihood of default in earlier yrs.



constant hazard rate

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Page 19
LOS c
- explain

Upfront Payment = PV(prot. leg) − $\underbrace{\text{PV(pr. leg)}}_{\text{with fixed rate}}$ (1% or 5%)

• if spread = rate, upfront payment = $0

∴ upfront payment = PV(spread) − PV(rate)
= (spread − rate) × Duration
= (credit spread − fixed coupon) × Duration

rearrange/

$$\text{credit spread} = \frac{\text{upfront payment}}{\text{Duration}} + \text{fixed coupon}$$

Price of CDS/100 par = 100 − upfront payment%

Page 20
LOS c
- explain

1/ high-yield company ⇒ 10-yr. credit spread = 600bps
5%

⇒ CDS Duration = 8 yrs.

$$\begin{matrix}\text{Upfront} \\ \text{Premium}\end{matrix} = \left(\begin{matrix}\text{Credit} \\ \text{Spread}\end{matrix} - \begin{matrix}\text{Fixed} \\ \text{Coupon}\end{matrix}\right) \times \text{Duration}$$

$$(600\text{bps} - 500\text{bps}) \times 8 = 8\%$$

2/ Sell 5-yr. protection ⇒ Investment grade company 1%

⇒ Upfront premium = 2% paid to buyer

⇒ CDS Duration = 4 yrs.

$$\begin{matrix}\text{Credit} \\ \text{Spread}\end{matrix} = \frac{\text{Upfront premium}}{\text{Duration}} + \text{Fixed Coupon}$$

$$= \frac{-.02}{4} + .01 = \quad 50\text{bps}$$

$$\begin{matrix}\text{Price of} \\ \text{CDS/100}\end{matrix} = 100 - \begin{matrix}\text{Upfront} \\ \text{premium}\end{matrix} = 100 - (-2\%) = 102\%$$

Page 21
LOS c
- explain

⇒ Valuation changes in CDS/

• PD, E(loss), shape of credit curve all change over time

$$\begin{matrix}\text{Profit for} \\ \text{CDS buyer}\end{matrix} \approx \underbrace{\text{changes in spread in bps} \times \text{Duration}}_{\text{\% Change in CDS price}} \times \text{Notional}$$

e.g./ Buy \$10M of 5-yr. protection

CDS Duration = 4 yrs.

$$\left.\begin{matrix} t_0 & \text{credit spread = 500 bps} \\ t_1 & \text{credit spread = 800 bps}\end{matrix}\right\} \text{widens}$$

∴ buyer benefits by:

$$\underbrace{300\text{bps} \times 4} \times \$10\text{M} = \$1.2\text{M}$$

% Price Change = 300bps × 4 = 12%

## Uses

Page 22
LOS d
- describe

⇒ facilitate the transfer of credit risk

• increase/decrease credit exposure

debtholder buys CDS to hedge (reduce) risk

seller adds credit exposure
- typically CDS dealers

- rather than buying a bond, an investor can sell a CDS

Buy a bond
- credit risk
- interest rate risk

Sell a CDS – credit risk only

• far less capital
• lower transaction costs
• CDS may be more liquid than the bond

Naked CDS
- buying without exposure

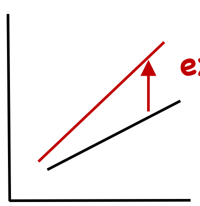
Page 23
LOS d
- describe

⇒ long/short positions

⇒ long in one CDS/short another (long/short trade)

e.g./ long CDX IG, short CDX HY

⇒ buy CDS of one maturity, sell CDS of another maturity



expect curve to steepen

long a short-term CDS (sell)

short a long-term CDS (buy)

Page 24
LOS e
- describe

⇒ amount of yield on a reference entity's bond attributable to credit risk should be the same as the credit spread on a CDS

- differences can result
  - opinion
  - models
  - liquidity

sets up a strategy known as a basis trade

e.g./ bond market implies 5% RP
CDS implies 4% RP
⇒ negative basis

- Buy the bond
- Buy the CDS

convergence captures the 1% differential

# REVIEW

# Term Structure/Interest Rate Dynamics

Review – 1

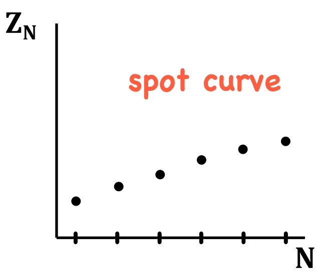
LOS a, b – describe/

spot rate – rate of interest for a single payment at a future date

forward rate – rate set today for a single payment security to be issued at some future date

– discount factor: $DF_N = {}^{1}/_{(1+Z_N)^N}$

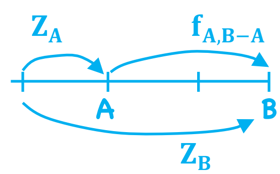
– the price of a single unit payment in N periods




– spot rates determined by forces of supply/demand

• forward pricing model

$$DF_B = DF_A \bullet F_{A,B-A}$$




• forward rate model

$$(1+Z_B)^B = (1+Z_A)^A\left(1+f_{A,B-A}\right)^{B-A}$$

Review – 2

LOS a, b – describe/

$$\left(\frac{1+Z_B}{1+Z_A}\right)^{A/B-A}\left[(1+Z_B) = \left(1+f_{A,B-A}\right)\right]$$

– if $Z_B > Z_A$ ➞ $f_{A,B-A} > Z_B$ ➞ forward curve lies above spot curve

– if $Z_B < Z_A$ ➞ $f_{A,B-A} < Z_B$ ➞ forward curve lies below spot curve

– par curve – YTMs on coupon paying bonds priced at par

– spot rates can be derived from par rates using a method called bootstrapping – solving the longest period spot rate from earliest maturing bond to longest

– relationship between spot rates and one-period forward rates

$$Z_T = \{\underbrace{(1+Z_1)(1+f_{1,1})(1+f_{2,1})\ldots(1+f_{T-1,1})}\}^{1/T-1}$$

geometric mean of Z, and a series of $T-1$ forward rates

Review - 3

LOS c - describe/

YTM - weighted-average of the spot rates used in the valuation of the bond (YTM = mwrr = bond's IRR)

- YTM = E(R) if
  a) held to maturity
  b) all CFs received
  c) all CFs reinvested at bond's YTM ∴ YTM = growth rate
  i.e. $PV(1+YTM)^T$ = future value

typically does not hold
- reinvestment rate ≠ YTM

- if spot curve is upward sloping, reinvestment is done at higher forward rates, ∴ E(R) > YTM

LOS d - describe/ - if forward rates are unbiased predictors of future spot rates → current spot curve is expected to evolve into the forward curve (i.e. current forward curve is the future spot curve)

Review - 4

LOS d - describe/

| spot evolves to forward curve | spot rises but not to the forward curve | spot rises above the forward curve |
|---|---|---|
| - all tenors will have the same return as a maturity matching bond | - buy longer than the investment horizon | - follow a maturity matching strategy or shorter |

rolling down the YC
- bond will be valued at successively lower yields
- the wider the spread, the longer the bond, the greater the total return

• E(R) = coupons + coupon reinvestment +/− cap. gain/loss

LOS e - explain/

swap rate - the rate on the fixed leg of an interest rate swap
- derived using s.t. interbank lending rates

Review - 5

LOS e - explain/ - yield curve of swap rates → swap curve

- swaps are priced such that $PV_{fl} = PV_{fx}$

- since $PV_{fl}$ = par = 1 ⇒ $PV_{fx}$ = 1

$$\text{swap rate}_N = \frac{1 - DF_N}{\sum_{i=1}^{N} DF_i}$$

- if gov't. bond market is not liquid > 1 yr. swap rates are a better benchmark of interest rates
- if private sector > public sector - swap curve more relevant measure of TVM
- both swap and spot curves used in FI valuation

LOS f - calculate/interpret/ swap spread = $\underbrace{\text{swap rate - 'on-the-run' gov't. rate}}_{\text{same maturity}}$

- gov't. rate → maturity premium (TVM)
- swap spread → credit and liquidity premiums

Review - 6

LOS g - describe/ Z-spread → a constant spread added to each spot rate → bond specific credit and liquidity risk

$$PV = \frac{PMT}{(1 + r_1 + Z)} + \frac{PMT}{(1 + r_2 + Z)^2} + \cdots + \frac{PMT + FV}{(1 + r_T + Z)}$$

I-spread → spread between a bond's YTM and an equal maturity swap rate

(equal maturity gov't. spot rate → G-spread)

- TED spread → T = T-Bill rate, ED - Libor (30-day)
  - both < 1 yr. → S.t. rates
  - good barometer of credit and liquidity risk

- Libor-OIS spread → $\underbrace{\text{unsecured loan rate - credit risk-free rate}}_{\text{same maturity}}$

overnight indexed swap rate

fx rate of a swap

$$[(1 + day_1)(1 + day_2)(1 + day_3) \dots (1 + day_t)]^{1/t} - 1$$

fl. rate.

Review - 7

LOS h - explain/describe/

- Expectations theory

1/ pure expectations - forward rates are unbiased predictors of future spot rate
∴ bonds of any maturity are perfect substitutes

2/ local expectations - E(R) ∀ bonds = $r_f$ over short time periods but not over longer periods

- Liquidity preference theory - liquidity premiums exist that increase with maturity
∴ predicts an upward sloping YC
- as such, forward rates as a prediction of future spot rates are biased upwards

- Segmented Markets Theory - each maturity segment can be thought of as a segmented market → market participants are unable or are unwilling to invest outside their maturity segment

Review - 8

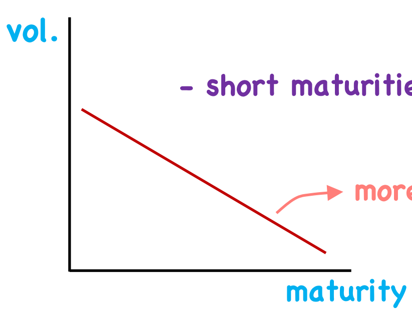
LOS h - explain/describe/

- Preferred Habitat - borrowers and lenders have a preferred maturity segment but will take advantage of compelling yields in other maturity segments

LOS i - explain/ shaping risk - the sensitivity of a bond's price to the changing shape of the YC

- factors affecting the shape of the YC
  - levels - dominant
  - steepness
  - curvature - smallest impact

LOS j - explain/




- short maturities - rates are more volatile - more strongly linked to monetary policy uncertainty

→ more strongly linked to uncertainty regarding the real economy and inflation

Review – 9

LOS j – explain/

– managing YC risk a) EffDur – parallel changes only

allow for mgmt. of shaping risk
b) Key rate duration – non-parallel changes
c) factor sensitivities

LOS k – explain/ short intermediate long

$^1/_3$ → inflation | monetary policy ← $^1/_3$

$^2/_3$ { GDP growth, monetary policy | inflation ← $^2/_3$

macro factors explaining variation in bond yields

Review – 10

LOS k – explain/

| | | Rates: Rise | Rates: Fall | |
|---|---|---|---|---|
| Curve | Steepens | bear steepener | bull steepener | bullet portfolio |
| | Flattens | bear flattener | bull flattener | barbell portfolio |

– rising rates are bearish for bond prices

– falling rates are bullish for bond prices

# Arbitrage-Free Valuation

Review - 1

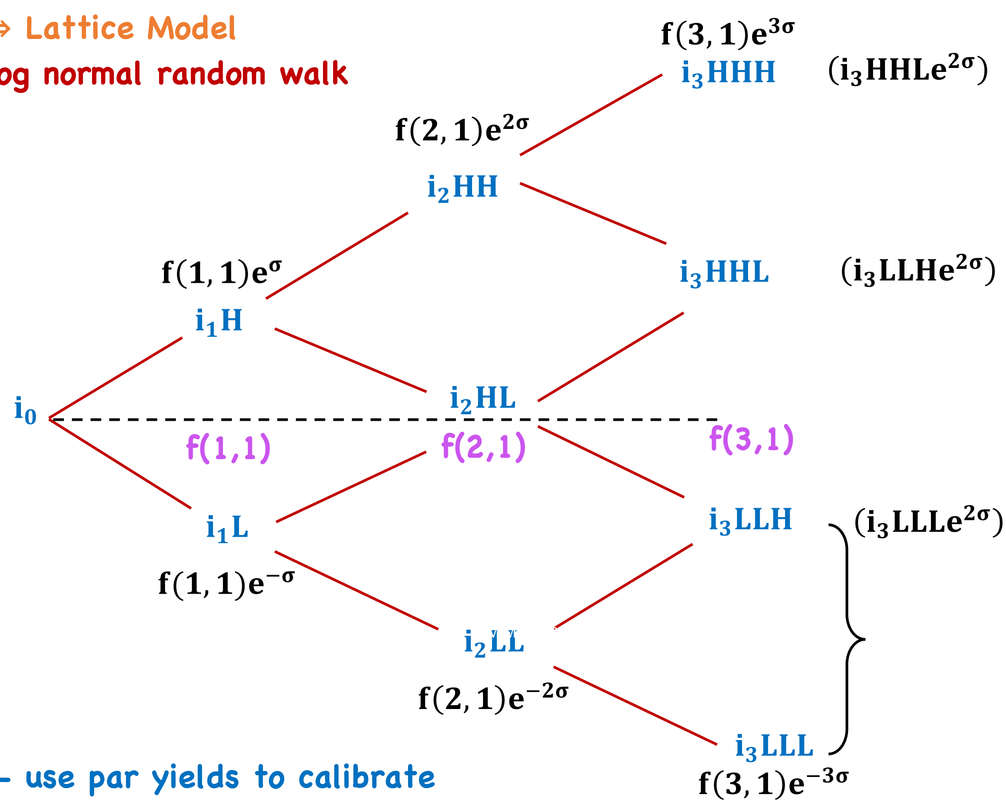
- arbitrage - riskless profit with zero investment
  1) Value Additivity - value of the whole should equal value of the parts
     95¢ for 1
     $95 for 105
  2) Dominance A: $100 → 105
     B: $200 → 220

⇒ Arbitrage Free Valuation/ an approach to security valuation that arrives at a price that is arbitrage free

⇒ Binomial Interest Rate Tree/
- option free + bonds w/ embedded options
- allows for volatility since complex bonds have cash flows that are interest rate dependent
- lognormal random process

Review - 2

⇒ Lattice Model
- log normal random walk

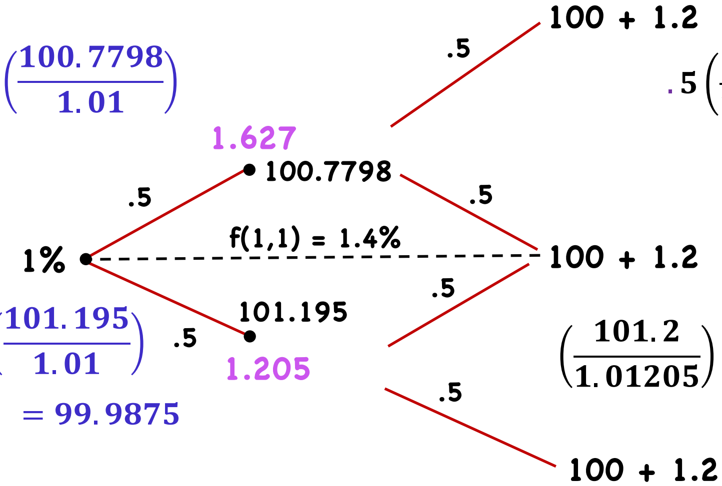



- higher volatility @ higher rates

- use par yields to calibrate the tree, one time-step at a time
  ⇒ Backward Induction

Review – 3

⇒ Lattice Model     2 yr. par = 1.2%     f(1,1) = 1.4%     σ = .15
r(1) = 1%




$$.5\left(\frac{100.7798}{1.01}\right)$$
$+$
$$.5\left(\frac{101.195}{1.01}\right)$$
$$= 99.9875$$

$$.5\left(\frac{101.2}{1.01627}\right) + .5\left(\frac{101.2}{1.01627}\right)$$
$$= 99.5798$$

$$\left(\frac{101.2}{1.01205}\right) = 99.995$$

$1.4e^{-\sigma} = 1.205 \times e^{2\sigma}$
$1.4e^{\sigma} =$

Review – 4

⇒ Pathwise Valuation/ – an alternative to backward induction

Pascal's Triangle

Discount a

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 yr. | | | | | 1 | | | | |
| 2 yr. | | | | 1 | | 1 | | | |
| 3 yr. | | | 1 | | 2 | | 1 | | |
| 4 yr. | | 1 | | 3 | | 3 | | 1 | |
| 5 yr. | 1 | | 4 | | 6 | | 4 | | 1 |

etc…

# of paths

- calculate PV of a bond for each possible path
  - take the average

⇒ Monte Carlo Simulation/ – used to simulate a sufficiently large # of interest rate paths

Review - 5

→ Term Structure Models/

**Arbitrage-free**

1/ Ho-Lee:

$$dr_t = \theta_t dt + \sigma dz$$

- time dependent drift term
- not mean reverting
- constant vol.
- neg. rates possible

2/ Kalotay-Williams-Fabozzi (KWF)

$$d\ln(r_t) = \theta_t dt + \sigma dz$$

log of $r_t$ → neg. rates not possible

**Equilibrium**

1/ Cox-Ingersoll-Ross (CIR)

$$dr_t = k(\theta - r_t)dt + \sigma\sqrt{r_t}dz$$

- mean reverting
- neg. rates not possible
- vol. varies with $\sqrt{r_t}$

2/ Vasicek model

$$dr_t = k(\theta - r_t)dt + \sigma dz$$

- mean reverting
- neg. rates possible
- constant vol.

# Valuation & Analysis

Review – 1

⇒ Embedded Option – contingency provision that can be exercised by the:

holder – put (holder value) issuer – call (issuer value)

depending on the course of interest rates

European – single date
American – anytime
Bermuda – several dates

} exercised typically at par

– convertible bond – call option on company's stock

$\underbrace{\text{Value of Callable Bond} = \text{Putable Bond} =}_{\text{Callable drops, Putable increases}} \underbrace{\text{Value of Straight Bond}}_{\text{unaffected}} \underbrace{- \text{Value of Call} \;\; + \text{Value of Put}}_{\text{value of option increases with interest rate volatility}}$

Callable drops
Putable increases

value of option increases with interest rate volatility

Review – 2

⇒ Interest Rates/

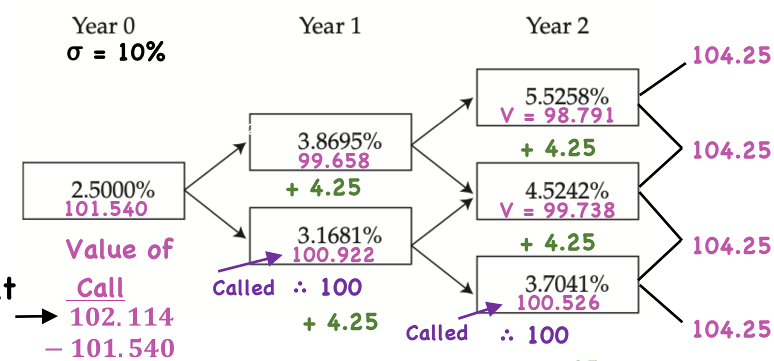
| Value of/ | Increase | | Decrease |
|---|---|---|---|
| St. Bond | decrease | | increase |
| Call Option | decrease | but at | increase |
| Callable Bond | decrease ← | slower rate → | increase |
| | | than straight | (cap) |
| Put Option | increase | but at a | decrease |
| Putable Bond | decrease ← | slower rate → | increase |
| | (floor) | than straight | |

• Upward sloping yield curve
  – less likely to be called
  – more likely to be put

⇒ Pricing on Option/

| Par | Spot | f |
|---|---|---|
| 2.5 | 2.5 | 2.5 |
| 3.0 | 3.008 | 3.518 |
| 3.5 | 3.524 | 4.564 |

3 yr., 4.25% Callable at par → $\sigma$ = 10%




Value of Call
102.114
− 101.540
0.574 vs. 0.407 @ $\sigma$ = 0%

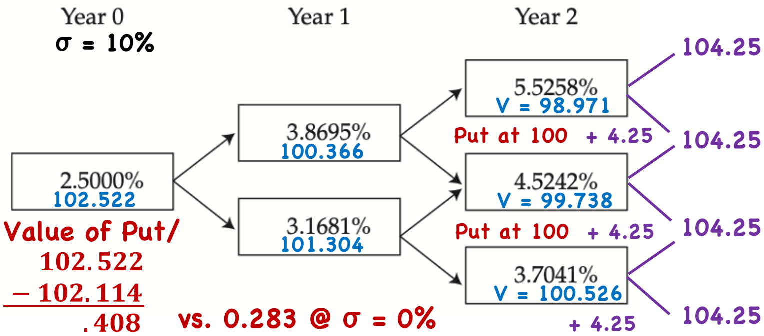
Review - 3

⇒ Pricing on Option/

| Par | Spot | f |
|---|---|---|
| 2.5 | 2.5 | 2.5 |
| 3.0 | 3.008 | 3.518 |
| 3.5 | 3.524 | 4.564 |

3 yr., 4.25% Putable at par

σ = 10%




Value of Put/
102.522
− 102.114
.408 vs. 0.283 @ σ = 0%

⇒ Option Adjusted Spread/ - for risky bonds

- add spread directly to the forward rates in the tree

2.50 + spread → 3.8695 + spread / 3.1681 + spread → etc...

Calibrate

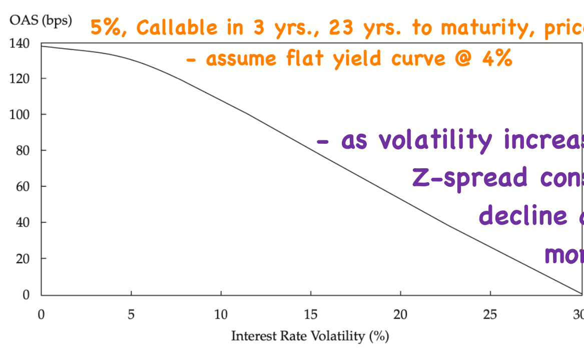
Review - 4

⇒ Option Adjusted Spread/ OAS is volatility dependent

Z-spread = OAS − option cost

5%, Callable in 3 yrs., 23 yrs. to maturity, priced @ 95

- assume flat yield curve @ 4%




- as volatility increases, holding the Z-spread constant, OAS would decline as the option grabs more of the spread

⇒ Duration

$$\text{EffDur} = \frac{(PV_-) - (PV_+)}{2\Delta \text{curve } PV_0}$$

1) Given $P_0$, find OAS for σ = x

2) shift curve down, generate new tree, revalue bond using OAS from 1)

3) repeat 2) for upward shift.

Review - 5

⇒ One-Sided/

$$1s - Dur = \frac{(PV_-) - PV_0}{2\Delta curve\ PV_0}$$

$$1s - Dur = \frac{PV_0 - (PV_+)}{2\Delta curve\ PV_0}$$

Call/
- more sensitive to interest rate increases

Put/ - more sensitive to interest rate decreases

⇒ Key Rate/ (partial duration)

$$\sum_{i=1}^{n} KeyRateDuration_i = EffDur$$

- for a par bond (option free) - duration is determined only by the rate matching the tenor of the bond

i.e. 10 yr. par - only affected by change in 10 yr. rate

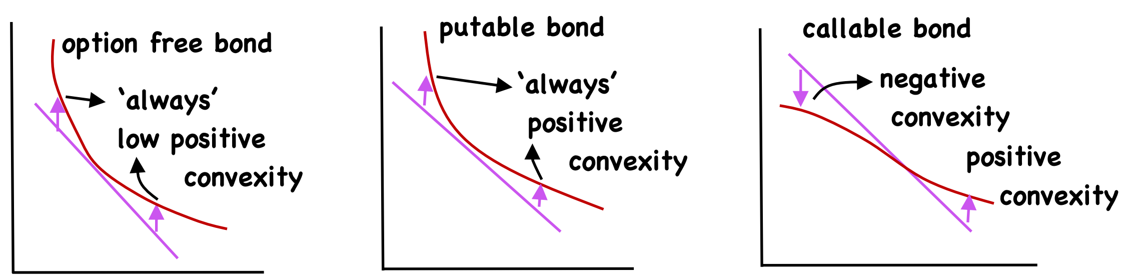
Callable - likely to be called
Putable - likely to be put
} duration(s) behave like option-free bonds = lockout period

e.g.: 30 yr. callable, 10 yr. lockout

Review - 6

⇒ Effective Convexity/

$$EffCon = \frac{(PV_-) + (PV_+) - 2PV_0}{(\Delta curve)^2\ PV_0}$$




⇒ Convertible Bonds/ straight bond + stock call option

(CR) conversion ratio = $\frac{\text{par}}{\text{conversion price}}$

(CV) conversion value = $P_0 \times CR$

minimum value of a con. B. = greater of
- CV
- value of underlying straight bond

Review - 7

⇒ Convertible Bonds/

market conversion premium/sh.

$$\text{MCP/sh.} = \frac{PV_0}{CR} - P_0 = \frac{\text{current price}}{\text{fixed conversion ratio}} - \text{current share price}$$

- even if $P_0$ > conversion price, if $\frac{PV_0}{CR} > P_0$, not worth converting.
- downside risk of Con. Bond/

$$\text{Premium over straight} = \frac{PV_0}{\text{Straight Value}} - 1$$

- upside potential/ linear with $P_0$

- Value of Con. B = Value of St. Bond + call option
  - issuer call option (forced conversion)
  + holder put option (if putable)

# Credit Analysis Models

Review – 1

LOS a – explain/

→ expected exposure – amount of money that could be lost in default without considering any recovery

→ loss given default – expected exposure × $(1 - \underbrace{\text{recovery rate}}_{\text{\%'age recovered in default}})$

→ probability of default – probability that an issuer will not meet its contractual obligations

→ credit valuation adjustment (CVA) – PV of the credit risk

| Date (1) | Exposure (2) | Recovery (3) | LGD (4) | POD (5) | POS (6) | Expected Loss (7) | DF (8) | PV of Expected Loss (9) |
|---|---|---|---|---|---|---|---|---|
| 0 | | | | | | | | |
| 1 | 88.8487 | 35.5395 | 53.3092 | 1.2500% | 98.7500% | 0.6664 | 0.970874 | 0.6470 |
| 2 | 91.5142 | 36.6057 | 54.9085 | 1.2344% | 97.5156% | 0.6778 | 0.942596 | 0.6389 |
| 3 | 94.2596 | 37.7038 | 56.5558 | 1.2189% | 96.2967% | 0.6894 | 0.915142 | 0.6309 |
| 4 | 97.0874 | 38.8350 | 58.2524 | 1.2037% | 95.0930% | 0.7012 | 0.888487 | 0.6230 |
| 5 | 100.0000 | 40.0000 | 60.0000 | 1.1887% | 93.9043% | 0.7132 | 0.862609 | 0.6152 |
| | | | | 6.0957% | | | CVA = | 3.1549 |

5-yr. zero (3% flat curve)

Review – 2

LOS a – describe/ PV(default free) – CVA = PV(risky bond)

credit spread = YTM(risky bond) – YTM(default free)

LOS b – explain/

- credit scores → retail lending market → ranks a borrower's credit riskiness, but does not provide a POD
- credit ratings → wholesale lending market → ranks credit risk of a company, government or ABS

IG / HY > 10 ratings each → issuer and issue

↓ typically for senior unsecured

– rating agency will issue a letter grade + outlook + 'watch' status

- pos./neg.
- stable

LOS c – calculate/

- transition matrix → can be used to estimate a 1-yr. rate of return given potential credit migration but no default

Review – 3

LOS c – calculate/

10 yr. – corporate bond

| From/To | AAA | AA | A | BBB | BB | B | CCC,CC,C | D |
|---|---|---|---|---|---|---|---|---|
| AAA | **90.00** | 9.00 | 0.60 | 0.15 | 0.10 | 0.10 | 0.05 | 0.00 |
| AA | 1.50 | **88.00** | 9.50 | 0.75 | 0.15 | 0.05 | 0.03 | 0.02 |
| A | 0.05 | 2.50 | **87.50** | 8.40 | 0.75 | 0.60 | 0.12 | 0.08 |
| BBB | 0.02 | 0.30 | 4.80 | **85.50** | 6.95 | 1.75 | 0.45 | 0.23 |
| BB | 0.01 | 0.06 | 0.30 | 7.75 | **79.50** | 8.75 | 2.38 | 1.25 |
| B | 0.00 | 0.05 | 0.15 | 1.40 | 9.15 | **76.60** | 8.45 | 4.20 |
| CCC,CC,C | 0.00 | 0.01 | 0.12 | 0.87 | 1.65 | 18.50 | **49.25** | 29.60 |
| Credit Spread | 0.60% | 0.90% | 1.10% | 1.50% | 3.40% | 6.50% | 9.50% | |

– assume ModDur of 7.2 at horizon end

– migrate from A → AA  $-7.2(.009 - .011) = +1.44\%$

A → B  $-7.2(.065 - .011) = -38.88\%$

$.0005(A \rightarrow AAA) + .025(A \rightarrow AA) + .875(0) + .084(A \rightarrow BBB) + \ldots + .012(A \rightarrow C)$

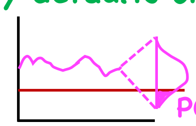
$= -.6342\%$

$\therefore E(R) = YTM - .6342\%$

Review – 4

LOS d – explain/

- structural model of credit risk – a company defaults on its debt if the value of A < D




$A_T = D_T + E_T$

$E_T = \max[A_T - k, 0]$ equity = call option on company's assets

$D_T = A_T - \max[A_T - k, 0]$

– assumes → company's assets are actively traded (w/ lognormal dist.)

→ default is endogenous (depends on structure of BS)

–/ → requires estimates of A and $\sigma_A$ (non-observable)

→ assumes D is as reported

- Reduced form model/ – default is exogeneous

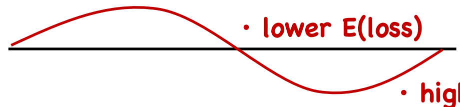
– default intensity varies depending on company performance and the business cycle

– assumes at least some of the company's debt trades

Review – 5

LOS d – explain/

- Reduced form model/
  - seeks to explain 'when' default occurs, not 'why'

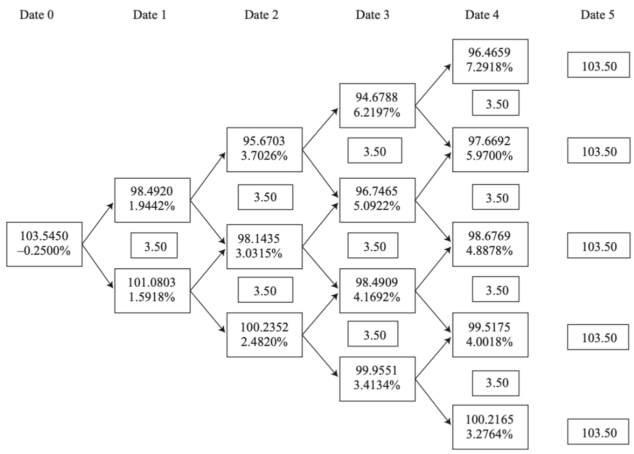


- lower POD
- lower E(loss)
- higher POD
- higher E(loss)

LOS e – calculate/

Step #1 – price risky bond assuming no default

| Date 2 | Date 3 |
|---|---|
| .25 | .125 |
| .50 | .375 |
| .25 | .375 |
| | .125 |




$.5^4 = .0625$
$\times$ (96.4659)
+
$.0625 \times 4 = .25$
$\times$ (97.6692)
+
$.0625 \times 6 = .375$
$\times$ (98.6769)
+
$.0625 \times 4 = .25$
$\times$ (99.5175)
+
$.0625 \times$ (100.2165)
+ 3.50

Review – 6

LOS e – calculate/

Step 2: CVA

| Date | Expected Exposure | LGD | POD | Discount Factor | CVA per Year |
|---|---|---|---|---|---|
| 0 | | | | | |
| 1 | 103.2862 | 61.9717 | 1.2500% | 1.002506 | 0.7766 |
| 2 | 101.5481 | 60.9289 | 1.2344% | 0.985093 | 0.7409 |
| 3 | 101.0433 | 60.6260 | 1.2189% | 0.955848 | 0.7064 |
| 4 | 102.0931 | 61.2559 | 1.2037% | 0.913225 | 0.6734 |
| 5 | 103.5000 | 62.1000 | 1.1887% | 0.870016 | 0.6422 |
| | | | 6.0957% | CVA = | 3.5394 |

x .6

Step 3: PV + YTM

103.5450 – 3.5394 = 100.0056
↓ DF ↓ CVA ↓ PV

PV = –100.0056
FV = 100
PMT = 3.5 CPT I/Y = .034988
N = 5

Step 4: Credit spread 3.4988 – 2.75% = .7488% → assumes spread is based entirely on credit risk

⇒ floating rate notes → main video clip

Page 14
LOS e
- calculate

- floating rate note → 5 yr., Index + .5% (QM)
→ exhibit 17

| Date | Expected Exposure | LGD | POD | Discount Factor | CVA per Year |
|---|---|---|---|---|---|
| 0 | | | | | |
| 1 | 102.1074 | 81.6859 | 0.5000% | 1.002506 | 0.4095 |
| 2 | 103.6583 | 82.9266 | 0.4975% | 0.985093 | 0.4064 |
| 3 | 104.4947 | 83.5957 | 0.4950% | 0.955848 | 0.3955 |
| 4 | ② 105.6535 | 95.0881 | 0.7388% | 0.913225 | 0.6416 |
| 5 | 105.4864 | 94.9377 | 0.7333% | 0.870016 | 0.6057 |
| | | | 2.9646% | CVA = | 2.4586 |

①

Changes
YR1 - YR3 = 80%
YR4 - YR5 = 90%

YR1 - YR3 = .5%
YR4 - YR5 = .75%

(.0625 × 107.7918) + (.25 × 106.47)
+ (.375 × 105.3878) + (.25 × 104.5018)
+ (.0625 × 103.7764) = 105.4864

② (.0625 × 100.4660)
+ (.25 × 100.4718)
+ (.375 × 100.4767)
+ (.25 × 100.4808)
+ (.0625) × 100.4841)

+ (.125 × 6.7197)
+ (.375 × 5.5922)
+ (.375 × 4.6692)
+ (.125 × 3.9134)

= 105.6535

Page 15
LOS e
- calculate

- floating rate note → 5 yr., Index + .5% (QM)
→ exhibit 17

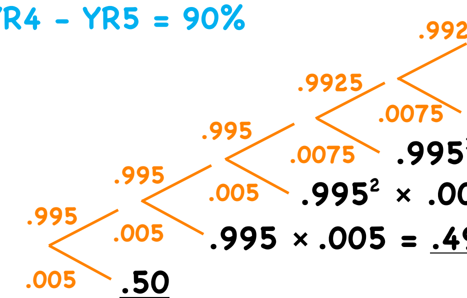
| Date | Expected Exposure | LGD | POD | Discount Factor | CVA per Year |
|---|---|---|---|---|---|
| 0 | | | | | |
| 1 | 102.1074 | 81.6859 | 0.5000% | 1.002506 | 0.4095 |
| 2 | 103.6583 | 82.9266 | 0.4975% | 0.985093 | 0.4064 |
| 3 | 104.4947 | 83.5957 | 0.4950% | 0.955848 | 0.3955 |
| 4 | 105.6535 | 95.0881 | 0.7388% | 0.913225 | 0.6416 |
| 5 | 105.4864 | 94.9377 | 0.7333% | 0.870016 | 0.6057 |
| | | | 2.9646% | CVA = | 2.4586 |

PV(ND) = 102.3633
$\sum$CVA − 2.4586
PV(D) 99.9047

∴ DM > QM
(discount bond)
(exhibit #19)

Changes
YR1 - YR3 = 80%
YR4 - YR5 = 90%

YR1 - YR3 = .5%
YR4 - YR5 = .75%




$.995^3 \times .9925 \times .0075 = \underline{.7333\%}$

$.995^3 \times .0075 = \underline{.7388\%}$

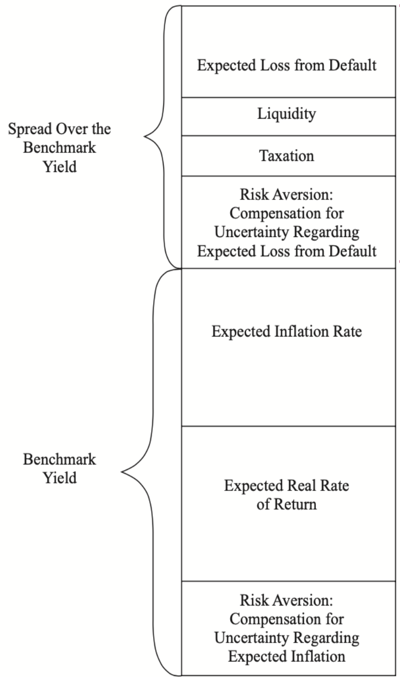
$.995^2 \times .005 = \underline{.4950\%}$

$.995 \times .005 = \underline{.4975\%}$

$\underline{.50}$

Review – 7

## LOS f – interpret/

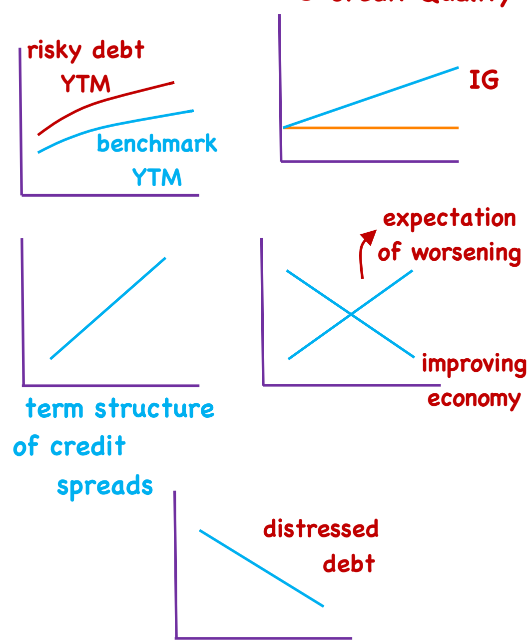



microeconomic factors (issuer & issue)

macroeconomic factors – affect all debt securities

## LOS g – explain

① Credit Quality




– as LGD & POD ↑, credit spreads increase

Review – 8

## LOS g – explain/

2. Financial conditions – credit risk affected by general economic conditions
3. Market demand/supply – credit curve is most heavily influenced by most frequently traded securities
   - frequently traded → narrow bid-ask spread
4. Company specific risk

## LOS h – compare

1/ homogeneous → conclusions from the class
heterogeneous → scrutiny on a loan-by-loan basis

2/ granularity: many → draw conclusions from summary statistics
few → analysis of each obligation

3/ origination & servicing → exposure to operational & counterparty risk

4/ structure of the secured debt transaction → the SPE + any structural enhancements

# Credit Default Swaps

Review – 1

⇒ CDS – underlying = borrower's credit quality

sell a CDS – long credit quality
buy a CDS – short credit quality

protection buyer → periodic PMTS → protection seller (premium leg)
protection buyer ← payment if default ← protection seller (protection leg)

⇒ Single-name CDS – a CDS on a $\underbrace{\text{specific borrower}}_{\text{reference entity}}$ and a specific $\underbrace{\text{debt obligation}}_{\text{reference obligation}}$ (usually senior unsecured)

– payoff determined by cheapest-to-deliver (same seniority)

⇒ Index CDS – a combination of borrowers (credit correlation)

Review – 2

⇒ Characteristics/
- notional amount
- expiration date
- periodic premium
  - 1% – inv. grade company/index
  - 5% – non-inv. grade

  rates are standardized
  – may require upfront payments

= Credit Events/
1) bankruptcy
2) failure to pay – on any outstanding obligation
3) restructuring

⇒ Settlement Protocols/ ~ 30 days after event
1) Physical settlement – holder sells bonds to CDS seller at par
2) Cash settlement payout ratio = 1 – recovery rate (CTD)

payment amount = payout ratio × Notional

Review - 3

⇒ Index CDS – $\underbrace{\text{CDX IG}}_{\text{equally weighted}}$ → 125 components

- when an entity defaults, removed from index, settled as single name CDS ($^1/_{125}$ of notional)
- index then moves forward w/ smaller notional

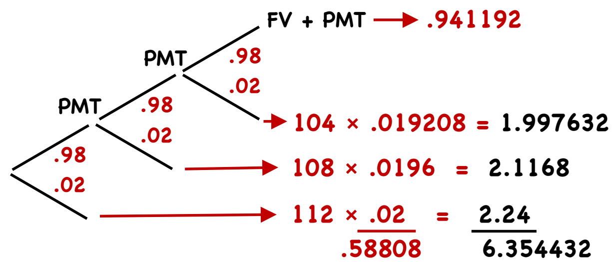
⇒ Valuation & Pricing/

- hazard rate – prob. of an event given it has not happened yet

PD = 1 – (prob. survival)$^t$ t = # of periods

E(loss) = PD × LGD

4%, 3 yr.




Review - 4

Upfront payment = PV(prot. leg) – PV(prem. leg) → 1% or 5%

> 0 buyer pays seller

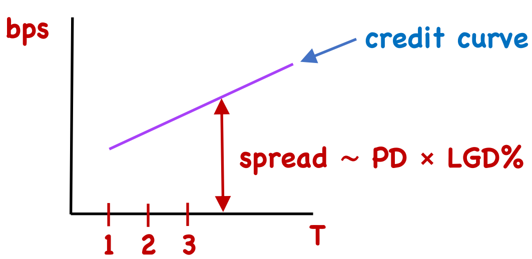
< 0 seller pays buyer

upfront PMT = PV(spread) – PV(rate)

= (spread – rate) × Duration

= (Credit spread – fixed coupon) × Duration

⇒ Term Structure of Credit Spreads/




upward – greater likelihood of default in later years

downward – " " earlier yrs.

flat – constant hazard rate

Review - 5

⇒ Profit for CDS buyer ≈ change in spread (bps) × Duration × Notional

⇒ Uses/ • increase or reduce credit exposure

Sell a CDS — hedging

- credit risk only (less capital, may be more liquid, lower trans. costs)

(buy a bond - interest rate + credit risk)

• Basis trade - bond market implies 5% RP

CDS implies 4% RP

- amt. of yield on a bond attributable to credit risk = credit spread on CDS

$\left.\begin{array}{l}\text{- buy bond}\\ \text{- buy CDS}\end{array}\right\}$ convergence captures 1%